14 Multiple Regression Analysis

THE MORTGAGE DEPARTMENT of the Bank of New England is studying data from recent loans. Of particular interest is how such factors as the value of the home being purchased, education level of the head of the household, age of the head of the household, current monthly mortgage payment, and gender of the head of the household relate to the family income. Are the proposed variables effective predictors of the dependent variable family income? (See the example/solution within the Review of Multiple Regression section.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO14-1 Use multiple regression analysis to describe and interpret a relationship between several independent variables and a dependent variable.

LO14-2 Evaluate how well a multiple regression equation fits the data.

LO14-3 Test hypotheses about the relationships inferred by a multiple regression model.

LO14-4 Evaluate the assumptions of multiple regression.

LO14-5 Use and interpret a qualitative, dummy variable in multiple regression.

LO14-6 Include and interpret an interaction effect in multiple regression analysis.

LO14-7 Apply stepwise regression to develop a multiple regression model.

LO14-8 Apply multiple regression techniques to develop a linear model.

INTRODUCTION

In Chapter 13, we described the relationship between a pair of interval- or ratio-scaled variables. We began the chapter by studying the correlation coefficient, which measures the strength of the relationship. A coefficient near plus or minus 1.00 (−.88 or .78, for example) indicates a very strong linear relationship, whereas a value near 0 (−.12 or .18, for example) indicates that the relationship is weak. Next we developed a procedure to determine a linear equation to express the relationship between the two variables. We referred to this as a *regression line.* This line describes the relationship between the variables. It also describes the overall pattern of a dependent variable (y) to a single independent or explanatory variable (x).

In multiple linear correlation and regression, we use additional independent variables (denoted $x_1, x_2, \ldots$, and so on) that help us better explain or predict the dependent variable (y). Almost all of the ideas we saw in simple linear correlation and regression extend to this more general situation. However, the additional independent variables do lead to some new considerations. Multiple regression analysis can be used either as a descriptive or as an inferential technique.

LO14-1

Use multiple regression analysis to describe and interpret a relationship between several independent variables and a dependent variable.

MULTIPLE REGRESSION ANALYSIS

The general descriptive form of a multiple linear equation is shown in formula (14–1). We use k to represent the number of independent variables. So k can be any positive integer.

GENERAL MULTIPLE REGRESSION EQUATION

$$\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + \ldots + b_kx_k \qquad \textbf{(14–1)}$$

where:

- a is the intercept, the value of $\hat{y}$ when all the X's are zero.
- b_j is the amount by which $\hat{y}$ changes when that particular x_j increases by one unit, with the values of all other independent variables held constant. The subscript j is simply a label that helps to identify each independent variable; it is not used in any calculations. Usually the subscript is an integer value between 1 and k, which is the number of independent variables. However, the subscript can also be a short or abbreviated label. For example, "age" could be used as a subscript to identify the independent variable, age.

In Chapter 13, the regression analysis described and tested the relationship between a dependent variable, $\hat{y}$ and a single independent variable, x. The relationship between ŷ and x was graphically portrayed by a line. When there are two independent variables, the regression equation is

$$\hat{y} = a + b_1x_1 + b_2x_2$$

Because there are two independent variables, this relationship is graphically portrayed as a plane and is shown in Chart 14–1. The chart shows the residuals as the difference between the actual y and the fitted $\hat{y}$ on the plane. If a multiple regression analysis includes more than two independent variables, we cannot use a graph to illustrate the analysis since graphs are limited to three dimensions.

To illustrate the interpretation of the intercept and the two regression coefficients, suppose the selling price of a home is directly related to the number of rooms and *inversely* related to its age. We let x_1 refer to the number of rooms, x_2 to the age of the home in years, and y to the selling price of the home in thousands of dollars ($000).

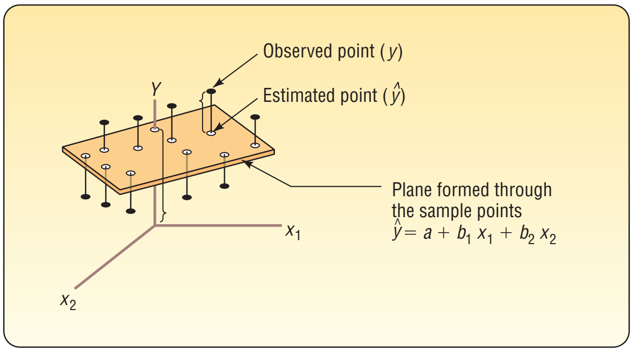


CHART 14–1 Regression Plane with 10 Sample Points

Suppose the regression equation, calculated using statistical software, is:

$$\hat{y} = 21.2 + 18.7x_1 - 0.25x_2$$

The intercept value of 21.2 indicates the regression equation (plane) intersects the *y*-axis at 21.2. This happens when both the number of rooms and the age of the home are zero. We could say that \$21,200 is the average value of a property without a house.

The first regression coefficient, 18.7, indicates that for each increase of one room in the size of a home, the selling price will increase by \$18.7 thousand (\$18,700), regardless of the age of the home. The second regression coefficient, −0.25, indicates that for each increase of one year in age, the selling price will *decrease* by \$.25 thousand (\$250), regardless of the number of rooms. As an example, a seven-room home that is 30 years old is expected to sell for \$144,600.

$$\hat{y} = 21.2 + 18.7x_1 - 0.25x_2 = 21.2 + 18.7(7) - 0.25(30) = 144.6$$

The values for the coefficients in the multiple linear equation are found by using the method of least squares. Recall from the previous chapter that the least squares method makes the sum of the squared differences between the fitted and actual values of y as small as possible, that is, the term $\Sigma(y - \hat{y})^2$ is minimized. The calculations are very tedious, so they are usually performed by a statistical software package.

In the following example, we show a multiple regression analysis using three independent variables employing Excel that produces a standard set of statistics and reports. Statistical software such as Minitab, MegaStat, and others provide advanced regression analysis techniques.

EXAMPLE

Salsberry Realty sells homes along the East Coast of the United States. One of the questions most frequently asked by prospective buyers is: If we purchase this home, how much can we expect to pay to heat it during the winter? The research department at Salsberry has been asked to develop some guidelines regarding heating costs for single-family homes. Three variables are thought to relate to the heating costs: (1) the mean daily outside temperature, (2) the number of inches of insulation in the attic, and (3) the age in years of the furnace. To investigate, Salsberry's research department selected a random sample of 20 recently sold homes. It determined the cost to heat each home last January, as well as the January outside temperature in the region, the number of inches of insulation in the attic, and the age of the furnace. The sample information is reported in Table 14–1.

TABLE 14–1 Factors in January Heating Cost for a Sample of 20 Homes

Home	Heating Cost ($)	Mean Outside Temperature (°F)	Attic Insulation (inches)	Age of Furnace (years)
1	$250	35	3	6
2	360	29	4	10
3	165	36	7	3
4	43	60	6	9
5	92	65	5	6
6	200	30	5	5
7	355	10	6	7
8	290	7	10	10
9	230	21	9	11
10	120	55	2	5
11	73	54	12	4
12	205	48	5	1
13	400	20	5	15
14	320	39	4	7
15	72	60	8	6
16	272	20	5	8
17	94	58	7	3
18	190	40	8	11
19	235	27	9	8
20	139	30	7	5

The data in Table 14–1 are available in Excel worksheet format at the textbook website, **www.mhhe.com/Lind17e**. The basic instructions for using Excel for these data are in the Software Commands in Appendix C.

Determine the multiple regression equation. Which variables are the independent variables? Which variable is the dependent variable? Discuss the regression coefficients. What does it indicate if some coefficients are positive and some coefficients are negative? What is the intercept value? What is the estimated heating cost for a home if the mean outside temperature is 30 degrees, there are 5 inches of insulation in the attic, and the furnace is 10 years old?

SOLUTION

We begin the analysis by defining the dependent and independent variables. The dependent variable is the January heating cost. It is represented by y. There are three independent variables:

- The mean outside temperature in January, represented by x_1.
- The number of inches of insulation in the attic, represented by x_2.
- The age in years of the furnace, represented by x_3.

Given these definitions, the general form of the multiple regression equation follows. The value $\hat{y}$ is used to estimate the value of y.

$$\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3$$

Now that we have defined the regression equation, we are ready to use Excel to compute all the statistics needed for the analysis. The output from Excel is shown on the following page.

To use the regression equation to predict the January heating cost, we need to know the values of the regression coefficients: b_1, b_2, and b_3. These are highlighted in the software reports. The software uses the variable names or labels associated with each independent variable. The regression equation intercept, *a,* is labeled “intercept” in the Excel output.

	A	B	C	D	E	F	G	H	I	J	K
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
3	360	29	4	10		*Regression Statistics*					
4	165	36	7	3		Multiple R	0.897				
5	43	60	6	9		R Square	0.804				
6	92	65	5	6		Adjusted R Square	0.767				
7	200	30	5	5		Standard Error	51.049				
8	355	10	6	7		Observations	20				
9	290	7	10	10							
10	230	21	9	11		ANOVA					
11	120	55	2	5			*df*	*SS*	*MS*	*F*	*Significance F*
12	73	54	12	4		Regression	3	171220.473	57073.491	21.901	0.000
13	205	48	5	1		Residual	16	41695.277	2605.955		
14	400	20	5	15		Total	19	212915.750			
15	320	39	4	7							
16	72	60	8	6			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	272	20	5	8		Intercept	427.194	59.601	7.168	0.000	
18	94	58	7	3		Temp	-4.583	0.772	-5.934	0.000	
19	190	40	8	11		Insul	-14.831	4.754	-3.119	0.007	
20	235	27	9	8		Age	6.101	4.012	1.521	0.148	
21	139	30	7	5							

In this case, the estimated regression equation is:

$$\hat{y} = 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3$$

We can now estimate or predict the January heating cost for a home if we know the mean outside temperature, the inches of insulation, and the age of the furnace. For an example home, the mean outside temperature for the month is 30 degrees (x_1), there are 5 inches of insulation in the attic (x_2), and the furnace is 10 years old (x_3). By substituting the values for the independent variables:

$$\hat{y} = 427.194 - 4.583(30) - 14.831(5) + 6.101(10) = 276.56$$

The estimated January heating cost is $276.56.

The regression coefficients, and their algebraic signs, also provide information about their individual relationships with the January heating cost. The regression coefficient for mean outside temperature is −4.583. The coefficient is negative and shows an inverse relationship between heating cost and temperature. This is not surprising. As the outside temperature increases, the cost to heat the home decreases. The numeric value of the regression coefficient provides more information. If the outside temperature increases by 1 degree and the other two independent variables remain constant, we can estimate a decrease of $4.583 in monthly heating cost. So if the mean temperature in Boston is 25 degrees and it is 35 degrees in Philadelphia, all other things being the same (insulation and age of furnace), we expect the heating cost would be $45.83 less in Philadelphia.

The attic insulation variable also shows an inverse relationship: the more insulation in the attic, the less the cost to heat the home. So the negative sign for this coefficient is logical. For each additional inch of insulation, we expect the cost to heat the home to decline $14.83 per month, holding the outside temperature and the age of the furnace constant.

The age of the furnace variable shows a direct relationship. With an older furnace, the cost to heat the home increases. Specifically, for each additional year older the furnace is, we expect the cost to increase $6.10 per month.

STATISTICS IN ACTION

Many studies indicate a woman will earn about 70% of what a man would for the same work. Researchers at the University of Michigan Institute for Social Research found that about one-third of the difference can be explained by such social factors as differences in education, seniority, and work interruptions. The remaining two-thirds is not explained by these social factors.

SELF-REVIEW 14–1

There are many restaurants in northeastern South Carolina. They serve beach vacationers in the summer, golfers in the fall and spring, and snowbirds in the winter. Bill and Joyce Tuneall manage several restaurants in the North Jersey area and are considering moving to Myrtle Beach, SC, to open a new restaurant. Before making a final decision, they wish to investigate existing restaurants and what variables seem to be related to profitability. They gather sample information where profit (reported in \$000) is the dependent variable and the independent variables are:

x_1 the number of parking spaces near the restaurant.
x_2 the number of hours the restaurant is open per week.
x_3 the distance from the SkyWheel, a landmark in Myrtle Beach.
x_4 the number of servers employed.
x_5 the number of years the current owner operated the restaurant.

The following is part of the output obtained using statistical software.

Predictor	Coefficient	SE Coefficient	t
Constant	2.50	1.50	1.667
x_1	3.00	1.50	2.000
x_2	4.00	3.00	1.333
x_3	-3.00	0.20	-15.000
x_4	0.20	0.05	4.000
x_5	1.00	1.50	0.667

(a) What is the amount of profit for a restaurant with 40 parking spaces that is open 72 hours per week, is 10 miles from the SkyWheel, has 20 servers, and has been operated by the current owner for 5 years?
(b) Interpret the values of b_2 and b_3 in the multiple regression equation.

EXERCISES

1. The director of marketing at Reeves Wholesale Products is studying monthly sales. Three independent variables were selected as estimators of sales: regional population, per capita income, and regional unemployment rate. The regression equation was computed to be (in dollars):

$$\hat{y} = 64{,}100 + 0.394x_1 + 9.6x_2 - 11{,}600x_3$$

a. What is the full name of the equation?
b. Interpret the number 64,100.
c. What are the estimated monthly sales for a particular region with a population of 796,000, per capita income of \$6,940, and an unemployment rate of 6.0%?

2. Thompson Photo Works purchased several new, highly sophisticated processing machines. The production department needed some guidance with respect to qualifications needed by an operator. Is age a factor? Is the length of service as an operator (in years) important? In order to explore further the factors needed to estimate performance on the new processing machines, four variables were listed:

x_1 = Length of time an employee was in the industry
x_2 = Mechanical aptitude test score
x_3 = Prior on-the-job rating
x_4 = Age

Performance on the new machine is designated *y*.

Thirty employees were selected at random. Data were collected for each, and their performances on the new machines were recorded. A few results are:

Name	Performance on New Machine, y	Length of Time in Industry, x_1	Mechanical Aptitude Score, x_2	Prior on-the-Job Performance, x_3	Age, x_4
Mike Miraglia	112	12	312	121	52
Sue Trythall	113	2	380	123	27

The equation is:

$$\hat{y} = 11.6 + 0.4x_1 + 0.286x_2 + 0.112x_3 + 0.002x_4$$

a. What is this equation called?
b. How many dependent variables are there? Independent variables?
c. What is the number 0.286 called?
d. As age increases by one year, how much does estimated performance on the new machine increase?
e. Carl Knox applied for a job at Photo Works. He has been in the business for 6 years and scored 280 on the mechanical aptitude test. Carl's prior on-the-job performance rating is 97, and he is 35 years old. Estimate Carl's performance on the new machine.

3. A consulting group was hired by the Human Resources Department at General Mills, Inc. to survey company employees regarding their degree of satisfaction with their quality of life. A special index, called the index of satisfaction, was used to measure satisfaction. Six factors were studied, namely, age at the time of first marriage (x_1), annual income (x_2), number of children living (x_3), value of all assets (x_4), status of health in the form of an index (x_5), and the average number of social activities per week—such as bowling and dancing (x_6). Suppose the multiple regression equation is:

$$\hat{y} = 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 + 0.19x_5 + 26.8x_6$$

a. What is the estimated index of satisfaction for a person who first married at 18, has an annual income of $26,500, has three children living, has assets of $156,000, has an index of health status of 141, and has 2.5 social activities a week on the average?
b. Which would add more to satisfaction, an additional income of $10,000 a year or two more social activities a week?

4. Cellulon, a manufacturer of home insulation, wants to develop guidelines for builders and consumers on how the thickness of the insulation in the attic of a home and the outdoor temperature affect natural gas consumption. In the laboratory, it varied the insulation thickness and temperature. A few of the findings are:

Monthly Natural Gas Consumption (cubic feet), y	Thickness of Insulation (inches), x_1	Outdoor Temperature (°F), x_2
30.3	6	40
26.9	12	40
22.1	8	49

On the basis of the sample results, the regression equation is:

$$\hat{y} = 62.65 - 1.86x_1 - 0.52x_2$$

a. How much natural gas can homeowners expect to use per month if they install 6 inches of insulation and the outdoor temperature is 40 degrees F?
b. What effect would installing 7 inches of insulation instead of 6 have on the monthly natural gas consumption (assuming the outdoor temperature remains at 40 degrees F)?
c. Why are the regression coefficients b_1 and b_2 negative? Is this logical?

LO14-2
Evaluate how well a multiple regression equation fits the data.

EVALUATING A MULTIPLE REGRESSION EQUATION

Many statistics and statistical methods are used to evaluate the relationship between a dependent variable and more than one independent variable. Our first step was to write the relationship in terms of a multiple regression equation. The next step follows on the concepts presented in Chapter 13 by using the information in an ANOVA table to evaluate how well the equation fits the data.

The ANOVA Table

As in Chapter 13, the statistical analysis of a multiple regression equation is summarized in an ANOVA table. To review, the total variation of the dependent variable, y, is divided into two components: (1) *regression,* or the variation of y explained by all the independent variables, and (2) *the error or residual,* or unexplained variation of y. These two categories are identified in the first column of an ANOVA table below. The column headed "*df*" refers to the degrees of freedom associated with each category. The total number of degrees of freedom is $n - 1$. The number of degrees of freedom in the regression is equal to the number of independent variables in the multiple regression equation. We call the regression degrees of freedom k. The number of degrees of freedom associated with the error term is equal to the total degrees of freedom, $n - 1$, minus the regression degrees of freedom, k. So, the residual or error degrees of freedom is $(n - 1) - k$, and is the same as $n - (k + 1)$.

Source	*df*	SS	MS	*F*
Regression	k	SSR	MSR = SSR/k	MSR/MSE
Residual or error	$n - (k + 1)$	SSE	MSE = SSE/$[n - (k + 1)]$	
Total	$n - 1$	SS total		

In the ANOVA table, the column headed, "SS", lists the sum of squares for each source of variation: regression, residual or error, and total. The sum of squares is the amount of variation attributable to each source.

The total variation of the dependent variable, y, is summarized in "SS total". You should note that this is simply the numerator of the usual formula to calculate any variation—in other words, the sum of the squared deviations from the mean. It is computed as:

$$\text{Total Sum of Squares} = \text{SS total} = \Sigma(y - \bar{y})^2$$

As we have seen, the total sum of squares is the sum of the regression and residual sum of squares. The regression sum of squares is the sum of the squared differences between the estimated or predicted values, $\hat{y}$ and the overall mean of y. The regression sum of squares is found by:

$$\text{Regression Sum of Squares} = \text{SSR} = \Sigma(\hat{y} - \bar{y})^2$$

The residual sum of squares is the sum of the squared differences between the observed values of the dependent variable, y, and their corresponding estimated or predicted values, $\hat{y}$. Notice that this difference is the error of estimating or predicting the dependent variable with the multiple regression equation. It is calculated as:

$$\text{Residual or Error Sum of Squares} = \text{SSE} = \Sigma(y - \hat{y})^2$$

We will use the ANOVA table information from the previous example to evaluate the regression equation to estimate January heating costs.

	A	B	C	D
1	Cost	Temp	Insul	Age
2	250	35	3	6
3	360	29	4	10
4	165	36	7	3
5	43	60	6	9
6	92	65	5	6
7	200	30	5	5
8	355	10	6	7
9	290	7	10	10
10	230	21	9	11
11	120	55	2	5
12	73	54	12	4
13	205	48	5	1
14	400	20	5	15
15	320	39	4	7
16	72	60	8	6
17	272	20	5	8
18	94	58	7	3
19	190	40	8	11
20	235	27	9	8

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.897
R Square	0.804
Adjusted R Square	0.767
Standard Error	51.049
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	171220.473	57073.491	21.901	0.000
Residual	16	41695.277	2605.955		
Total	19	212915.750			

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	427.194	59.601	7.168	0.000
Temp	-4.583	0.772	-5.934	0.000
Insul	-14.831	4.754	-3.119	0.007
Age	6.101	4.012	1.521	0.148

Multiple Standard Error of Estimate

We begin with the **multiple standard error of estimate.** Recall that the standard error of estimate is comparable to the standard deviation. To explain the details of the standard error of estimate, refer to the first sampled home in row 2 in the Excel spreadsheet on the previous page. The actual heating cost for the first observation, y, is \$250; the outside temperature, x_1, is 35 degrees; the depth of insulation, x_2, is 3 inches; and the age of the furnace, x_3, is 6 years. Using the regression equation developed in the previous section, the estimated heating cost for this home is:

$$\begin{aligned}\hat{y} &= 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3 \\ &= 427.194 - 4.583(35) - 14.831(3) + 6.101(6) \\ &= 258.90\end{aligned}$$

So we would estimate that a home with a mean January outside temperature of 35 degrees, 3 inches of insulation, and a 6-year-old furnace would cost \$258.90 to heat. The actual heating cost was \$250, so the residual—which is the difference between the actual value and the estimated value—is $y - \hat{y} = 250 - 258.90 = -8.90$. This difference of \$8.90 is the random or unexplained error for the first home sampled. Our next step is to square this difference—that is, find $(y - \hat{y})^2 = (250 - 258.90)^2 = (-8.90)^2 = 79.21$.

If we repeat this calculation for the other 19 observations and sum all 20 squared differences, the total will be the residual or error sum of squares from the ANOVA table. Using this information, we can calculate the multiple standard error of the estimate as:

MULTIPLE STANDARD ERROR OF ESTIMATE

$$s_{y \cdot 123 \ldots k} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - (k + 1)}} = \sqrt{\frac{SSE}{n - (k + 1)}} \qquad \textbf{(14–2)}$$

where:

y is the actual observation.
$\hat{y}$ is the estimated value computed from the regression equation.
n is the number of observations in the sample.
k is the number of independent variables.
SSE is the Residual Sum of Squares from an ANOVA table.

There is more information in the ANOVA table that can be used to compute the multiple standard error of estimate. The column headed "MS" reports the mean squares for the regression and residual variation. These values are calculated as the sum of squares divided by the corresponding degrees of freedom. The multiple standard error of estimate is equal to the square root of the residual MS, which is also called the mean square error or the MSE.

$$s_{y \cdot 123 \ldots k} = \sqrt{\text{MSE}} = \sqrt{2605.995} = \$51.05$$

How do we interpret the standard error of estimate of 51.05? It is the typical "error" when we use this equation to predict the cost. First, the units are the same as the dependent variable, so the standard error is in dollars, \$51.05. Second, we expect the residuals to be approximately normally distributed, so about 68% of the residuals will be within $\pm$\$51.05 and about 95% within $\pm 2(51.05)$, or $\pm$\$102.10. As before with similar measures of dispersion, such as the standard error of estimate in Chapter 13, a smaller multiple standard error indicates a better or more effective predictive equation.

Coefficient of Multiple Determination

Next, let's look at the coefficient of multiple determination. Recall from the previous chapter the coefficient of determination is defined as the percent of variation in the dependent variable explained, or accounted for, by the independent variable. In the multiple regression case, we extend this definition as follows.

COEFFICIENT OF MULTIPLE DETERMINATION The percent of variation in the dependent variable, y, explained by the set of independent variables, $x_1, x_2, x_3, \ldots x_k$.

The characteristics of the coefficient of multiple determination are:

1. **It is symbolized by a capital *R* squared.** In other words, it is written as R^2 because it is calculated as the square of a correlation coefficient.
2. **It can range from 0 to 1.** A value near 0 indicates little association between the set of independent variables and the dependent variable. A value near 1 means a strong association.
3. **It cannot assume negative values.** Any number that is squared or raised to the second power cannot be negative.
4. **It is easy to interpret.** Because R^2 is a value between 0 and 1, it is easy to interpret, compare, and understand.

We can calculate the coefficient of determination from the information found in the ANOVA table. We look in the sum of squares column, which is labeled SS in the Excel output, and use the regression sum of squares, SSR, then divide by the total sum of squares, SS total.

COEFFICIENT OF MULTIPLE DETERMINATION

$$R^2 = \frac{\text{SSR}}{\text{SS total}} \qquad \textbf{(14–3)}$$

We can use the regression and the total sum of squares from the ANOVA table highlighted in the Excel output appearing earlier in this section and compute the coefficient of determination.

$$R^2 = \frac{\text{SSR}}{\text{SS total}} = \frac{171{,}220.473}{212{,}915.750} = .804$$

How do we interpret this value? We conclude that the independent variables (outside temperature, amount of insulation, and age of furnace) explain, or account for, 80.4% of the variation in heating cost. To put it another way, 19.6% of the variation is due to other sources, such as random error or variables not included in the analysis. Using the ANOVA table, 19.6% is the error sum of squares divided by the total sum of squares. Knowing that the SSR + SSE = SS total, the following relationship is true.

$$1 - R^2 = 1 - \frac{\text{SSR}}{\text{SS total}} = \frac{\text{SSE}}{\text{SS total}} = \frac{41{,}695.277}{212{,}915.750} = .196$$

Adjusted Coefficient of Determination

The coefficient of determination tends to increase as more independent variables are added to the multiple regression model. Each new independent variable causes the predictions to be more accurate. That, in turn, makes SSE smaller and SSR larger. Hence, R^2 increases only because the total number of independent variables increases and not because the added independent variable is a good predictor of the dependent variable. In fact, if the number of variables, k, and the sample size, n, are equal, the coefficient of determination is 1.0. In practice, this situation is rare and would also be ethically questionable. To balance the effect that the number of independent variables has on the coefficient of multiple determination, statistical software packages use an *adjusted* coefficient of multiple determination.

ADJUSTED COEFFICIENT OF DETERMINATION

$$R^2_{\text{adj}} = 1 - \frac{\dfrac{\text{SSE}}{n - (k + 1)}}{\dfrac{\text{SS total}}{n - 1}} \qquad \textbf{(14–4)}$$

The error and total sum of squares are divided by their degrees of freedom. Notice especially the degrees of freedom for the error sum of squares include k, the number of independent variables. For the cost of heating example, the adjusted coefficient of determination is:

$$R^2_{\text{adj}} = 1 - \frac{\dfrac{41{,}695.277}{20 - (3 + 1)}}{\dfrac{212{,}915.750}{20 - 1}} = 1 - \frac{2{,}605.955}{11{,}206.092} = 1 - .233 = .767$$

If we compare the R^2 (0.80) to the adjusted R^2 (0.767), the difference in this case is small.

SELF-REVIEW 14–2

Refer to Self-Review 14–1 on the subject of restaurants in Myrtle Beach. The ANOVA portion of the regression output is presented below.

```
Analysis of Variance
Source            DF     SS    MS
Regression         5    100    20
Residual Error    20     40     2
Total             25    140
```

(a) How large was the sample?
(b) How many independent variables are there?
(c) How many dependent variables are there?
(d) Compute the standard error of estimate. About 95% of the residuals will be between what two values?
(e) Determine the coefficient of multiple determination. Interpret this value.
(f) Find the coefficient of multiple determination, adjusted for the degrees of freedom.

EXERCISES

5. Consider the ANOVA table that follows.

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	77.907	38.954	4.14	0.021
Residual Error	62	583.693	9.414		
Total	64	661.600			

a. Determine the standard error of estimate. About 95% of the residuals will be between what two values?
b. Determine the coefficient of multiple determination. Interpret this value.
c. Determine the coefficient of multiple determination, adjusted for the degrees of freedom.

6. Consider the ANOVA table that follows.

Analysis of Variance

Source	DF	SS	MS	F
Regression	5	3710.00	742.00	12.89
Residual Error	46	2647.38	57.55	
Total	51	6357.38		

a. Determine the standard error of estimate. About 95% of the residuals will be between what two values?
b. Determine the coefficient of multiple determination. Interpret this value.
c. Determine the coefficient of multiple determination, adjusted for the degrees of freedom.

LO14-3
Test hypotheses about the relationships inferred by a multiple regression model.

INFERENCES IN MULTIPLE LINEAR REGRESSION

Thus far, multiple regression analysis has been viewed only as a way to describe the relationship between a dependent variable and several independent variables. However, the least squares method also has the ability to draw inferences or generalizations about the relationship for an entire population. Recall that when you create confidence intervals or perform hypothesis tests as a part of inferential statistics, you view the data as a random sample taken from some population.

In the multiple regression setting, we assume there is an unknown population regression equation that relates the dependent variable to the k independent variables. This is sometimes called a **model** of the relationship. In symbols we write:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

This equation is analogous to formula (14–1) except the coefficients are now reported as Greek letters. We use the Greek letters to denote *population parameters*. Then under a certain set of assumptions, which will be discussed shortly, the computed values of a and b_i are sample statistics. These sample statistics are point estimates of the corresponding population parameters α and β_i. For example, the sample regression coefficient b_2 is a point estimate of the population parameter β_2. The sampling distribution of these point estimates follows the normal probability distribution. These sampling distributions are each centered at their respective parameter values. To put it another way, the means of the sampling distributions are equal to the parameter values to be estimated. Thus, by using the properties of the sampling distributions of these statistics, inferences about the population parameters are possible.

Global Test: Testing the Multiple Regression Model

We can test the ability of the independent variables $X_1, X_2, \ldots, X_k$ to explain the behavior of the dependent variable Y. To put this in question form: Can the dependent variable be estimated without relying on the independent variables? The test used is referred to as the **global test.** Basically, it investigates whether it is possible that all the independent variables have zero regression coefficients.

To relate this question to the heating cost example, we will test whether the three independent variables (amount of insulation in the attic, mean daily outside temperature, and age of furnace) effectively estimate home heating costs. In testing the hypothesis, we first state the null hypothesis and the alternate hypothesis in terms of the three population parameters, β_1, β_2, and β_3 . Recall that b_1, b_2, and b_3 are sample regression coefficients and are not used in the hypothesis statements. In the null hypothesis, we test whether the regression coefficients in the population are all zero. The null hypothesis is:

$$H_0\text{: } \beta_1 = \beta_2 = \beta_3 = 0$$

The alternate hypothesis is:

$$H_1\text{: Not all the } \beta_i\text{'s are 0.}$$

If the hypothesis test fails to reject the null hypothesis, it implies the regression coefficients are all zero and, logically, are of no value in estimating the dependent variable (heating cost). Should that be the case, we would have to search for some other independent variables—or take a different approach—to predict home heating costs.

To test the null hypothesis that the multiple regression coefficients are all zero, we employ the F distribution introduced in Chapter 12. We will use the .05 level of significance. Recall these characteristics of the F distribution:

1. **There is a family of *F* distributions.** Each time the degrees of freedom in either the numerator or the denominator change, a new F distribution is created.
2. **The *F* distribution cannot be negative.** The smallest possible value is 0.
3. **It is a continuous distribution.** The distribution can assume an infinite number of values between 0 and positive infinity.
4. **It is positively skewed.** The long tail of the distribution is to the right-hand side. As the number of degrees of freedom increases in both the numerator and the denominator, the distribution approaches the normal probability distribution. That is, the distribution will move toward a symmetric distribution.
5. **It is asymptotic.** As the values of X increase, the F curve will approach the horizontal axis, but will never touch it.

The F-statistic to test the global hypothesis follows. As in Chapter 12, it is the ratio of two variances. In this case, the numerator is the regression sum of squares divided by its degrees of freedom, k. The denominator is the residual sum of squares divided by its degrees of freedom, $n - (k + 1)$. The formula follows.

GLOBAL TEST
$$F = \frac{\text{SSR}/k}{\text{SSE}/[n - (k + 1)]} \quad \textbf{(14–5)}$$

Using the values from the ANOVA table on page 492, the F-statistic is

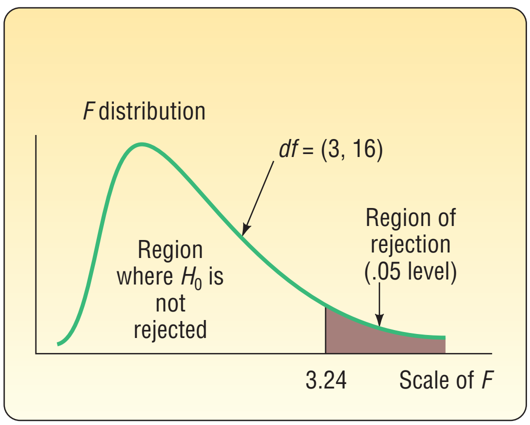
$$F = \frac{\text{SSR}/k}{\text{SSE}/[n - (k + 1)]} = \frac{171{,}220.473/3}{41{,}695.277/[20 - (3 + 1)]} = 21.90$$

Remember that the F-statistic tests the basic null hypothesis that two variances or, in this case, two mean squares are equal. In our global multiple regression hypothesis test, we will reject the null hypothesis, H_0, that all regression coefficients are zero when the regression mean square is larger in comparison to the residual mean square. If this is true, the F-statistic will be relatively large and in the far right tail of the F distribution, and the p-value will be small, that is, less than our choice of significance level of 0.05. Thus, we will reject the null hypothesis.

As with other hypothesis-testing methods, the decision rule can be based on either of two methods: (1) comparing the test statistic to a critical value or (2) calculating a p-value based on the test statistic and comparing the p-value to the significance level. The critical value method using the F-statistic requires three pieces of information: (1) the numerator degrees of freedom, (2) the denominator degrees of freedom, and (3) the significance level. The degrees of freedom for the numerator and the denominator are reported in the Excel ANOVA table that follows. The ANOVA output is highlighted in light green. The top number in the column marked "*df*" is 3, indicating there are 3 degrees of freedom in the numerator. This value corresponds to the number of independent variables. The middle number in the "*df*" column (16) indicates that there are 16 degrees of freedom in the denominator. The number 16 is found by $n - (k - 1) = 20 - (3 - 1) = 16$.

	A	B	C	D	G	H	I	J	K	L	M
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
3	360	29	4	10		*Regression Statistics*					
4	165	36	7	3		Multiple R	0.897				
5	43	60	6	9		R Square	0.804				
6	92	65	5	6		Adjusted R Square	0.767				
7	200	30	5	5		Standard Error	51.049				
8	355	10	6	7		Observations	20				
9	290	7	10	10							
10	230	21	9	11		ANOVA					
11	120	55	2	5			*df*	*SS*	*MS*	*F*	*Significance F*
12	73	54	12	4		Regression	3	171220.473	57073.491	21.901	0.000
13	205	48	5	1		Residual	16	41695.277	2605.955		
14	400	20	5	15		Total	19	212915.750			
15	320	39	4	7							
16	72	60	8	6			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	272	20	5	8		Intercept	427.194	59.601	7.168	0.000	
18	94	58	7	3		Temp	-4.583	0.772	-5.934	0.000	
19	190	40	8	11		Insul	-14.831	4.754	-3.119	0.007	
20	235	27	9	8		Age	6.101	4.012	1.521	0.148	

The critical value of F is found in Appendix B.6. Using the table for the .05 significance level, move horizontally to 3 degrees of freedom in the numerator, then down to 16 degrees of freedom in the denominator, and read the critical value. It is 3.24. The region where H_0 is not rejected and the region where H_0 is rejected are shown in the following diagram.



Continuing with the global test, the decision rule is: Do not reject the null hypothesis, H_0, that all the regression coefficients are 0 if the computed value of F is less than or equal to 3.24. If the computed F is greater than 3.24, reject H_0 and accept the alternate hypothesis, H_1.

The computed value of F is 21.90, which is in the rejection region. The null hypothesis that all the multiple regression coefficients are zero is therefore rejected. This means that at least one of the independent variables has the ability to explain the variation in the dependent variable (heating cost). We expected this decision. Logically, the outside temperature, the amount of insulation, or the age of the furnace has a great bearing on heating costs. The global test assures us that they do.

Testing the null hypothesis can also be based on a *p*-value, which is reported in the statistical software output for all hypothesis tests. In the case of the *F*-statistic, the *p*-value is defined as the probability of observing an *F*-value as large or larger than the *F* test statistic, assuming the null hypothesis is true. If the *p*-value is less than our selected significance level, then we decide to reject the null hypothesis. The ANOVA shows the *F*-statistic's *p*-value is equal to 0.000. It is clearly less than our significance level of 0.05. Therefore, we decide to reject the global null hypothesis and conclude that at least one of the regression coefficients is not equal to zero.

Evaluating Individual Regression Coefficients

So far we have shown that at least one, but not necessarily all, of the regression coefficients is not equal to zero and thus useful for predictions. The next step is to test the independent variables *individually* to determine which regression coefficients may be 0 and which are not.

Why is it important to know if any of the β_i's equal 0? If a β could equal 0, it implies that this particular independent variable is of no value in explaining any variation in the dependent value. If there are coefficients for which H_0 cannot be rejected, we may want to eliminate them from the regression equation.

Our strategy is to use three sets of hypotheses: one for temperature, one for insulation, and one for age of the furnace.

For temperature:	For insulation:	For furnace age:
$H_0: \beta_1 = 0$	$H_0: \beta_2 = 0$	$H_0: \beta_3 = 0$
$H_1: \beta_1 \neq 0$	$H_1: \beta_2 \neq 0$	$H_1: \beta_3 \neq 0$

We will test the hypotheses at the .05 level. Note that these are two-tailed tests.

The test statistic follows Student's *t* distribution with $n - (k + 1)$ degrees of freedom. The number of sample observations is *n*. There are 20 homes in the study, so $n = 20$. The number of independent variables is *k*, which is 3. Thus, there are $n - (k + 1) = 20 - (3 + 1) = 16$ degrees of freedom.

The critical value for *t* is in Appendix B.5. For a two-tailed test with 16 degrees of freedom using the .05 significance level, H_0 is rejected if *t* is less than −2.120 or greater than 2.120.

Refer to the Excel output earlier in this section. The column highlighted in orange, headed Coefficients, shows the values for the multiple regression equation:

$$\hat{y} = 427.194 - 4.583x_1 - 14.831x_2 + 6.101x_3$$

Interpreting the term $-4.583x_1$ in the equation: For each degree increase in temperature, we predict that heating cost will decrease $4.58, holding the insulation and age of the furnace variables constant.

The column in the Excel output labeled "Standard Error" shows the standard error of the sample regression coefficients. Recall that Salsberry Realty selected a sample of 20 homes along the East Coast of the United States. If Salsberry Realty selected a second random sample and computed the regression coefficients for that sample, the values would not be exactly the same. If the sampling process was repeated many times, we could construct a sampling distribution for each of these regression coefficients. The column labeled "Standard Error" estimates the variability for each of these regression coefficients. The sampling distributions of the coefficients follow the *t* distribution with $n - (k + 1)$ degrees of freedom. Hence, we are able to test the independent variables individually to determine whether the regression coefficients differ from zero. The formula is:

TESTING INDIVIDUAL REGRESSION COEFFICIENTS

$$t = \frac{b_i - 0}{s_{b_i}} \qquad \textbf{(14–6)}$$

The b_i refers to any one of the regression coefficients, and s_{b_i} refers to the standard deviation of that distribution of the regression coefficient. We include 0 in the equation because the null hypothesis is $\beta_i = 0$.

To illustrate this formula, refer to the test of the regression coefficient for the independent variable temperature. From the output earlier in this section, the regression coefficient for temperature is −4.583. The standard deviation of the sampling distribution of the regression coefficient for the independent variable temperature is 0.772. Inserting these values in formula (14–6):

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{-4.583 - 0}{0.772} = -5.937$$

The computed value of t is −5.937 for temperature (the small difference between the computed value and that shown on the Excel output is due to rounding) and −3.119 for insulation. Both of these t-values are in the rejection region to the left of −2.120. Thus, we conclude that the regression coefficients for the temperature and insulation variables are *not* zero. The computed t for the age of the furnace is 1.521, so we conclude that the coefficient could equal 0. The independent variable age of the furnace is not a significant predictor of heating cost. The results of these hypothesis tests indicate that the analysis should focus on temperature and insulation as predictors of heating cost.

We can also use p-values to test the individual regression coefficients. Again, these are commonly reported in statistical software output. The computed value of t for temperature on the Excel output is −5.934 and has a p-value of 0.000. Because the p-value is less than 0.05, the regression coefficient for the independent variable temperature is not equal to zero and should be included in the equation to predict heating costs. For insulation, the value of t is −3.119 and has a p-value of 0.007. As with temperature, the p-value is less than 0.05, so we conclude that the insulation regression coefficient is not equal to zero and should be included in the equation to predict heating cost. In contrast to temperature and insulation, the p-value to test the "age of the furnace" regression coefficient is 0.148. It is clearly greater than 0.05, so we conclude that the "age of furnace" regression coefficient could equal 0. Further, as an independent variable it is not a significant predictor of heating cost. Thus, age of furnace should not be included in the equation to predict heating costs.

At this point, we need to develop a strategy for deleting independent variables. In the Salsberry Realty case, there were three independent variables. For the age of the furnace variable, we failed to reject the null hypothesis that the regression coefficient was zero. It is clear that we should drop that variable and rerun the regression equation. Below is the Excel output where heating cost is the dependent variable and outside temperature and amount of insulation are the independent variables.

	A	B	C	D	E	F	G	H	I	J	K
1	Cost	Temp	Insul	Age		SUMMARY OUTPUT					
2	250	35	3	6							
3	360	29	4	10		*Regression Statistics*					
4	165	36	7	3		Multiple R	0.881				
5	43	60	6	9		R Square	0.776				
6	92	65	5	6		Adjusted R Square	0.749				
7	200	30	5	5		Standard Error	52.982				
8	355	10	6	7		Observations	20				
9	290	7	10	10							
10	230	21	9	11		ANOVA					
11	120	55	2	5			*df*	*SS*	*MS*	*F*	*Significance F*
12	73	54	12	4		Regression	2	165194.521	82597.261	29.424	0.000
13	205	48	5	1		Residual	17	47721.229	2807.131		
14	400	20	5	15		Total	19	212915.750			
15	320	39	4	7							
16	72	60	8	6			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	272	20	5	8		Intercept	490.286	44.410	11.040	0.000	
18	94	58	7	3		Temp	-5.150	0.702	-7.337	0.000	
19	190	40	8	11		Insul	-14.718	4.934	-2.983	0.008	
20	235	27	9	8							

Summarizing the results from this new output:

1. The new regression equation is:

$$\hat{y} = 490.286 - 5.150x_1 - 14.718x_2$$

Notice that the regression coefficients for outside temperature (x_1) and amount of insulation (x_2) are similar to but not exactly the same as when we included the independent variable age of the furnace. Compare the above equation to that in the Excel output earlier in this section. Both of the regression coefficients are negative as in the earlier equation.

2. The details of the global test are as follows:

$$H_0 : \beta_1 = \beta_2 = 0$$
$$H_1 : \text{Not all of the } \beta_i\text{'s} = 0$$

The F distribution is the test statistic and there are $k = 2$ degrees of freedom in the numerator and $n - (k + 1) = 20 - (2 + 1) = 17$ degrees of freedom in the denominator. Using the .05 significance level and Appendix B.6, the decision rule is to reject H_0 if F is greater than 3.59. We compute the value of F as follows:

$$F = \frac{\text{SSR}/k}{\text{SSE}/[n - (k + 1)]} = \frac{165{,}194.521/2}{47{,}721.229/[20 - (2 + 1)]} = 29.424$$

Because the computed value of F (29.424) is greater than the critical value (3.59), the null hypothesis is rejected and the alternate accepted. We conclude that at least one of the regression coefficients is different from 0.

Using the p-value, the F test statistic (29.424) has a p-value (0.000), which is clearly less than 0.05. Therefore, we reject the null hypothesis and accept the alternate. We conclude that at least one of the regression coefficients is different from 0.

3. The next step is to conduct a test of the regression coefficients individually. We want to determine if one or both of the regression coefficients are different from 0. The null and alternate hypotheses for each of the independent variables are:

Outside Temperature	Insulation
$H_0: \beta_1 = 0$	$H_0: \beta_2 = 0$
$H_1: \beta_1 \neq 0$	$H_1: \beta_2 \neq 0$

The test statistic is the t distribution with $n - (k + 1) = 20 - (2 + 1) = 17$ degrees of freedom. Using the .05 significance level and Appendix B.5, the decision rule is to reject H_0 if the computed value of t is less than -2.110 or greater than 2.110.

Outside Temperature	Insulation
$t = \frac{b_1 - 0}{s_{b_1}} = \frac{-5.150 - 0}{0.702} = -7.337$	$t = \frac{b_2 - 0}{s_{b_2}} = \frac{-14.718 - 0}{4.934} = -2.983$

In both tests, we reject H_0 and accept H_1. We conclude that each of the regression coefficients is different from 0. Both outside temperature and amount of insulation are useful variables in explaining the variation in heating costs.

Using p-values, the p-value for the temperature t-statistic is 0.000 and the p-value for the insulation t-statistic is 0.008. Both p-values are less than 0.05, so in both tests we reject the null hypothesis and conclude that each of the regression coefficients is different from 0. Both outside temperature and amount of insulation are useful variables in explaining the variation in heating costs.

In the heating cost example, it was clear which independent variable to delete. However, in some instances which variable to delete may not be as clear-cut. To explain, suppose we develop a multiple regression equation based on five independent variables. We conduct the global test and find that some of the regression coefficients are different from zero. Next, we test the regression coefficients individually and find that three are significant and two are not. The preferred procedure is to drop the single independent variable with the *smallest absolute* t *value* or *largest* p-*value* and rerun the regression equation with the four remaining variables, then, on the new regression equation with four independent variables, conduct the individual tests. If there are still regression coefficients that are not significant, again drop the variable with the smallest absolute *t* value or the largest, nonsignificant *p*-value. To describe the process in another way, we should delete only one variable at a time. Each time we delete a variable, we need to rerun the regression equation and check the remaining variables.

This process of selecting variables to include in a regression model can be automated, using Excel, Minitab, MegaStat, or other statistical software. Most of the software systems include methods to sequentially remove and/or add independent variables and at the same time provide estimates of the percentage of variation explained (the *R*-square term). Two of the common methods are **stepwise regression** and **best subset regression.** It may take a long time, but in the extreme we could compute every regression between the dependent variable and any possible subset of the independent variables.

Unfortunately, on occasion, the software may work "too hard" to find an equation that fits all the quirks of your particular data set. The suggested equation may not represent the relationship in the population. Judgment is needed to choose among the equations presented. Consider whether the results are logical. They should have a simple interpretation and be consistent with your knowledge of the application under study.

SELF-REVIEW 14–3

The regression output about eating places in Myrtle Beach is repeated below (see earlier self-reviews).

Predictor	Coefficient	SE Coefficient	t	p-value	
Constant	2.50	1.50	1.667	0.111	
x_1	3.00	1.50	2.000	0.056	
x_2	4.00	3.00	1.333	0.194	
x_3	-3.00	0.20	-15.000	0.000	
x_4	0.20	0.05	4.000	0.000	
x_5	1.00	1.50	0.667	0.511	
Analysis of Variance					
Source	DF	SS	MS	F	p-value
Regression	5	100	20	10	0.000
Residual Error	20	40	2		
Total	25	140			

(a) Perform a global test of hypothesis to check if any of the regression coefficients are different from 0. What do you decide? Use the .05 significance level.

(b) Do an individual test of each independent variable. Which variables would you consider eliminating? Use the .05 significance level.

(c) Outline a plan for possibly removing independent variables.

EXERCISES

7. Given the following regression output,

Predictor	Coefficient	SE Coefficient	t	p-value	
Constant	84.998	1.863	45.62	0.000	
x_1	2.391	1.200	1.99	0.051	
x_2	-0.409	0.172	-2.38	0.021	
Analysis of Variance					
Source	DF	SS	MS	F	p-value
Regression	2	77.907	38.954	4.138	0.021
Residual Error	62	583.693	9.414		
Total	64	661.600			

answer the following questions:

a. Write the regression equation.

b. If x_1 is 4 and x_2 is 11, what is the expected or predicted value of the dependent variable?

c. How large is the sample? How many independent variables are there?

d. Conduct a global test of hypothesis to see if any of the set of regression coefficients could be different from 0. Use the .05 significance level. What is your conclusion?

e. Conduct a test of hypothesis for each independent variable. Use the .05 significance level. Which variable would you consider eliminating?

f. Outline a strategy for deleting independent variables in this case.

8. The following regression output was obtained from a study of architectural firms. The dependent variable is the total amount of fees in millions of dollars.

Predictor	Coefficient	SE Coefficient	t	p-value	
Constant	7.987	2.967	2.690	0.010	
x_1	0.122	0.031	3.920	0.000	
x_2	-1.220	0.053	-2.270	0.028	
x_3	-0.063	0.039	-1.610	0.114	
x_4	0.523	0.142	3.690	0.001	
x_5	-0.065	0.040	-1.620	0.112	
Analysis of Variance					
Source	DF	SS	MS	F	p-value
Regression	5	371000	742	12.89	0.000
Residual Error	46	2647.38	57.55		
Total	51	6357.38			

x_1 is the number of architects employed by the company.
x_2 is the number of engineers employed by the company.
x_3 is the number of years involved with health care projects.
x_4 is the number of states in which the firm operates.
x_5 is the percent of the firm's work that is health care–related.

a. Write out the regression equation.

b. How large is the sample? How many independent variables are there?

c. Conduct a global test of hypothesis to see if any of the set of regression coefficients could be different from 0. Use the .05 significance level. What is your conclusion?

d. Conduct a test of hypothesis for each independent variable. Use the .05 significance level. Which variable would you consider eliminating first?

e. Outline a strategy for deleting independent variables in this case.

LO14-4
Evaluate the assumptions of multiple regression.

EVALUATING THE ASSUMPTIONS OF MULTIPLE REGRESSION

In the previous section, we described the methods to statistically evaluate the multiple regression equation. The results of the test let us know if at least one of the coefficients was not equal to zero and we described a procedure of evaluating each regression coefficient. We also discussed the decision-making process for including and excluding independent variables in the multiple regression equation.

It is important to know that the validity of the statistical global and individual tests rely on several assumptions. So if the assumptions are not true, the results might be biased or misleading. However, strict adherence to the following assumptions is not always possible. Fortunately, the statistical techniques discussed in this chapter are robust enough to work effectively even when one or more of the assumptions are violated. Even if the values in the multiple regression equation are "off" slightly, our estimates using a multiple regression equation will be closer than any that could be made otherwise.

In Chapter 13, we listed the necessary assumptions for regression when we considered only a single independent variable. The assumptions for multiple regression are similar.

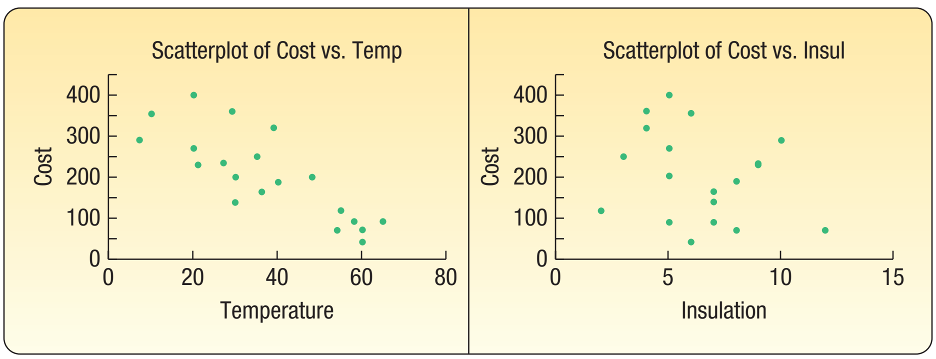
1. **There is a linear relationship.** That is, there is a straight-line relationship between the dependent variable and the set of independent variables.
2. **The variation in the residuals is the same for both large and small values of $\hat{y}$.** To put it another way, $(y - \hat{y})$ is unrelated to whether $\hat{y}$ is large or small.
3. **The residuals follow the normal probability distribution.** Recall the residual is the difference between the actual value of y and the estimated value $\hat{y}$. So the term $(y - \hat{y})$ is computed for every observation in the data set. These residuals should approximately follow a normal probability distribution with a mean of 0.
4. **The independent variables should not be correlated.** That is, we would like to select a set of independent variables that are not themselves correlated.
5. **The residuals are independent.** This means that successive observations of the dependent variable are not correlated. This assumption is often violated when time is involved with the sampled observations.

In this section, we present a brief discussion of each of these assumptions. In addition, we provide methods to validate these assumptions and indicate the consequences if these assumptions cannot be met. For those interested in additional discussion, search on the term "Applied Linear Models."

Linear Relationship

Let's begin with the linearity assumption. The idea is that the relationship between the set of independent variables and the dependent variable is linear. If we are considering two independent variables, we can visualize this assumption. The two independent variables and the dependent variable would form a three-dimensional space. The regression equation would then form a plane as shown on page 490. We can evaluate this assumption with scatter diagrams and residual plots.

Using Scatter Diagrams The evaluation of a multiple regression equation should always include a scatter diagram that plots the dependent variable against each independent variable. These graphs help us to visualize the relationships and provide some initial information about the direction (positive or negative), linearity, and strength of the relationship. For example, the scatter diagrams for the home heating example follow. The plots suggest a fairly strong negative, linear relationship between heating cost and temperature, and a negative relationship between heating cost and insulation.

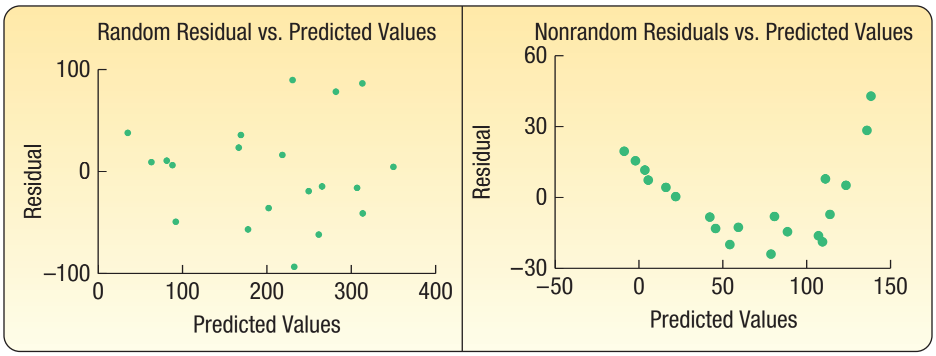


Using Residual Plots Recall that a residual $(y - \hat{y})$ can be computed using the multiple regression equation for each observation in a data set. In Chapter 13, we discussed the idea that the best regression line passed through the center of the data in a scatter plot. In this case, you would find a good number of the observations above the regression line (these residuals would have a positive sign) and a good number of the observations below the line (these residuals would have a negative sign). Further, the observations would be scattered above and below the line over the entire range of the independent variable.

The same concept is true for multiple regression, but we cannot graphically portray the multiple regression. However, plots of the residuals can help us evaluate the linearity of the multiple regression equation. To investigate, the residuals are plotted on the vertical axis against the predicted variable, $\hat{y}$. In the following graphs, the left graph shows the residual plots for the home heating cost example. Notice the following:

- The residuals are plotted on the vertical axis and are centered around zero. There are both positive and negative residuals.
- The residual plots show a random distribution of positive and negative values across the entire range of the variable plotted on the horizontal axis.
- The points are scattered and there is no obvious pattern, so there is no reason to doubt the linearity assumption.

The plot on the right shows nonrandom residuals. See that the residual plot does *not* show a random distribution of positive and negative values across the entire range of the variable plotted on the horizontal axis. In fact, the graph shows a non-linear pattern of the residuals. This indicates the relationship is probably not linear. In this case, we would evaluate different transformations of the variables in the equation as discussed in Chapter 13.



Variation in Residuals Same for Large and Small $\hat{y}$ Values

This requirement indicates that the variation in the residuals is constant, regardless of whether the predicted values are large or small. To cite a specific example which may violate the assumption, suppose we use the single independent variable age to explain variation in monthly income. We suspect that as age increases so does income, but it also seems reasonable that as age increases there may be more variation around the regression line. That is, there will likely be more variation in income for 50-year-olds than for 35-year-olds. The requirement for constant variation around the regression line is called **homoscedasticity.**

HOMOSCEDASTICITY The variation around the regression equation is the same for all of the values of the independent variables.

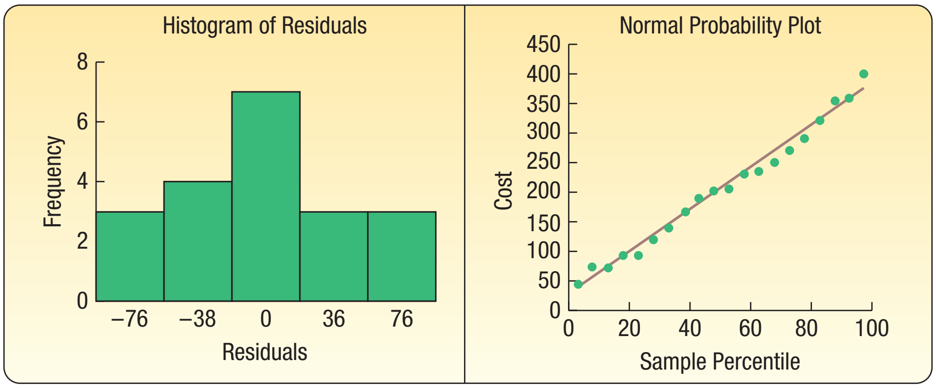
To check for homoscedasticity, the residuals are plotted against $\hat{y}$. This is the same graph that we used to evaluate the assumption of linearity. Based on the scatter diagram, it is reasonable to conclude that this assumption has not been violated.

Distribution of Residuals

To be sure that the inferences we make in the global and individual hypothesis tests are valid, we evaluate the distribution of residuals. Ideally, the residuals should follow a normal probability distribution.

To evaluate this assumption, we can organize the residuals into a frequency distribution. The Histogram of Residuals graph is shown on the left for the home heating cost example. Although it is difficult to show that the residuals follow a normal distribution with only 20 observations, it does appear the normality assumption is reasonable.

Another graph that helps to evaluate the assumption of normally distributed residuals is called a Normal Probability Plot and is shown to the right of the histogram. This graphical analysis is often included in statistical software. If the plotted points are fairly close to a straight line drawn from the lower left to the upper right of the graph, the normal probability plot supports the assumption of normally distributed residuals This plot supports the assumption of normally distributed residuals.



In this case, both graphs support the assumption that the residuals follow the normal probability distribution. Therefore, the inferences that we made based on the global and individual hypothesis tests are supported with the results of this evaluation.

Multicollinearity

Multicollinearity exists when independent variables are correlated. Correlated independent variables make it difficult to make inferences about the individual regression coefficients and their individual effects on the dependent variable. In practice, it is nearly impossible to select variables that are completely unrelated. To put it another way, it is nearly impossible to create a set of independent variables that are not correlated to some degree. However, a general understanding of the issue of multicollinearity is important.

First, multicollinearity does not affect a multiple regression equation's ability to predict the dependent variable. However, when we are interested in evaluating the relationship between each independent variable and the dependent variable, multicollinearity may show unexpected results.

For example, if we use two highly correlated independent variables, high school GPA and high school class rank, to predict the GPA of incoming college freshmen (dependent variable), we would expect that both independent variables would be positively related to the dependent variable. However, because the independent variables are highly correlated, one of the independent variables may have an unexpected and inexplicable negative sign. In essence, these two independent variables are redundant in that they explain the same variation in the dependent variable.

A second reason to avoid correlated independent variables is they may lead to erroneous results in the hypothesis tests for the individual independent variables. This is due to the instability of the standard error of estimate. Several clues that indicate problems with multicollinearity include the following:

1. An independent variable known to be an important predictor ends up having a regression coefficient that is not significant.
2. A regression coefficient that should have a positive sign turns out to be negative, or vice versa.
3. When an independent variable is added or removed, there is a drastic change in the values of the remaining regression coefficients.

In our evaluation of a multiple regression equation, an approach to reducing the effects of multicollinearity is to carefully select the independent variables that are included in the regression equation. A general rule is if the correlation between two independent variables is between −0.70 and 0.70, there likely is not a problem using both of the independent variables. A more precise test is to use the **variance inflation factor.** It is usually written *VIF*. The value of *VIF* is found as follows:

VARIANCE INFLATION FACTOR

$$VIF = \frac{1}{1 - R_j^2} \qquad \textbf{(14–7)}$$

The term R_j^2 refers to the coefficient of determination, where the selected *independent variable* is used as a dependent variable and the remaining independent variables are used as independent variables. A *VIF* greater than 10 is considered unsatisfactory, indicating that the independent variable should be removed from the analysis. The following example will explain the details of finding the *VIF*.

EXAMPLE

Refer to the data in Table 14–1, which relate the heating cost to the independent variables: outside temperature, amount of insulation, and age of furnace. Develop a correlation matrix for all the independent variables. Does it appear there is a problem with multicollinearity? Find and interpret the variance inflation factor for each of the independent variables.

SOLUTION

We begin by finding the correlation matrix for the dependent variable and the three independent variables. A correlation matrix shows the correlation between all pairs of the variables. A portion of that output follows:

	Cost	*Temp*	*Insul*	*Age*
Cost	1.000			
Temp	–0.812	1.000		
Insul	–0.257	–0.103	1.000	
Age	0.537	–0.486	0.064	1.000

The highlighted area indicates the correlation among the independent variables. Because all of the correlations are between −.70 and .70, we do not suspect problems with multicollinearity. The largest correlation among the independent variables is −0.486 between age and temperature.

To confirm this conclusion, we compute the *VIF* for each of the three independent variables. We will consider the independent variable temperature first. We use the Regression Analysis in Excel to find the multiple coefficient of determination with temperature as the *dependent variable* and amount of insulation and age of the furnace as independent variables. The relevant regression output follows.

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.491				
R Square	0.241				
Adjusted R Square	0.152				
Standard Error	16.031				
Observations	20				
ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	1390.291	695.145	2.705	0.096
Residual	17	4368.909	256.995		
Total	19	5759.200			

The coefficient of determination is .241, so inserting this value into the *VIF* formula:

$$VIF = \frac{1}{1 - R_1^2} = \frac{1}{1 - .241} = 1.32$$

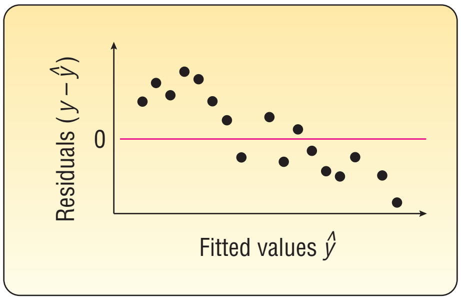
The *VIF* value of 1.32 is less than the upper limit of 10. This indicates that the independent variable temperature is not strongly correlated with the other independent variables.

Again, to find the *VIF* for insulation we would develop a regression equation with insulation as the *dependent variable* and temperature and age of furnace as independent variables. For this equation, the R^2 is .011 and, using formula (14–7), the *VIF* for insulation would be 1.011. To find the *VIF* for age, we would develop a regression equation with age as the dependent variable and temperature and insulation as the independent variables. For this equation, the R^2 is .236 and, using formula (14–7), the *VIF* for age would be 1.310. All the *VIF* values are less than 10. Hence, we conclude there is not a problem with multicollinearity in this example.

Independent Observations

The fifth assumption about regression and correlation analysis is that successive residuals should be independent. This means that there is not a pattern to the residuals, the residuals are not highly correlated, and there are not long runs of positive or negative residuals. When successive residuals are correlated, we refer to this condition as **autocorrelation.**

Autocorrelation frequently occurs when the data are collected over a period of time. For example, we wish to predict yearly sales of Agis Software Inc. based on the time and the amount spent on advertising. The dependent variable is yearly sales and the independent variables are time and amount spent on advertising. It is likely that for a period of time the actual points will be above the regression plane (remember there are two independent variables) and then for a period of time the points will be below the regression plane. The graph below shows the residuals plotted on the vertical axis and the fitted values $\hat{y}$ on the horizontal axis. Note the run of residuals above the mean of the residuals, followed by a run below the mean. A scatter plot such as this would indicate possible autocorrelation.



There is a test for autocorrelation, called the Durbin-Watson. It is discussed in Chapter 18.

LO14-5
Use and interpret a qualitative, dummy variable in multiple regression.

QUALITATIVE INDEPENDENT VARIABLES

In the previous example/solution regarding heating cost, the two independent variables outside temperature and insulation were quantitative; that is, numerical in nature. Frequently we wish to use nominal-scale variables—such as gender, whether the home has a swimming pool, or whether the sports team was the home or the visiting team—in our analysis. These are called **qualitative variables** because they describe a particular quality or attribute, such as gender measured as male or female. To use a qualitative variable in regression analysis, we use a scheme of **dummy variables** in which one of the two possible conditions is coded 0 and the other 1.

STATISTICS IN ACTION

Multiple regression has been used in a variety of legal proceedings. It is particularly useful in cases alleging discrimination by gender or race. As an example, suppose that a woman alleges that Company X's wage rates are unfair to women. To support the claim, the plaintiff produces data showing that, on the average, women earn less than men. In response, Company X argues that its wage rates are based on experience, training, and skill and that its female employees, on the average, are younger and less experienced than the male employees. In fact, the company might further argue that the current situation is actually due to its recent successful efforts to hire more women.

DUMMY VARIABLE A variable in which there are only two possible outcomes. For analysis, one of the outcomes is coded a 1 and the other a 0.

For example, we are interested in estimating an executive's salary on the basis of years of job experience and whether he or she graduated from college. "Graduation from college" can take on only one of two conditions: yes or no. Thus, it is considered a qualitative variable.

Suppose in the Salsberry Realty example that the independent variable "garage" is added. For those homes without an attached garage, 0 is used; for homes with an attached garage, a 1 is used. We will refer to the "garage" variable as x_4. The data from Table 14–2 are entered into the Excel system. Recall that the variable "age of the furnace" is not included in the analysis because we determined that it was not significantly related to heating cost.

The output from Excel is:

	A	B	C	D	E	F	G	H	I	J	K
1	Cost	Temp	Insul	Garage		SUMMARY OUTPUT					
2	250	35	3	0							
3	360	29	4	1		*Regression Statistics*					
4	165	36	7	0		Multiple R	0.933				
5	43	60	6	0		R Square	0.870				
6	92	65	5	0		Adjusted R Square	0.845				
7	200	30	5	0		Standard Error	41.618				
8	355	10	6	1		Observations	20				
9	290	7	10	1							
10	230	21	9	0		ANOVA					
11	120	55	2	0			*df*	*SS*	*MS*	*F*	*Significance F*
12	73	54	12	0		Regression	3	185202.269	61734.090	35.641	0.000
13	205	48	5	1		Residual	16	27713.481	1732.093		
14	400	20	5	1		Total	19	212915.750			
15	320	39	4	1							
16	72	60	8	0			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	272	20	5	1		Intercept	393.666	45.001	8.748	0.000	
18	94	58	7	0		Temp	-3.963	0.653	-6.072	0.000	
19	190	40	8	1		Insul	-11.334	4.002	-2.832	0.012	
20	235	27	9	0		Garage	77.432	22.783	3.399	0.004	

TABLE 14–2 Home Heating Costs, Temperature, Insulation, and Presence of a Garage for a Sample of 20 Homes

Cost, y	Temperature, x_1	Insulation, x_2	Garage, x_4
$250	35	3	0
360	29	4	1
165	36	7	0
43	60	6	0
92	65	5	0
200	30	5	0
355	10	6	1
290	7	10	1
230	21	9	0
120	55	2	0
73	54	12	0
205	48	5	1
400	20	5	1
320	39	4	1
72	60	8	0
272	20	5	1
94	58	7	0
190	40	8	1
235	27	9	0
139	30	7	0

What is the effect of the garage variable? Should it be included in the analysis? To show the effect of the variable, suppose we have two homes exactly alike next to each other in Buffalo, New York; one has an attached garage and the other does not. Both homes have 3 inches of insulation, and the mean January temperature in Buffalo is 20 degrees. For the house without an attached garage, a 0 is substituted for x_4 in the regression equation. The estimated heating cost is $280.404, found by:

$$\begin{aligned}\hat{y} &= 393.666 - 3.963x_1 - 11.334x_2 + 77.432x_4 \\ &= 393.666 - 3.963(20) - 11.334(3) + 77.432(0) = 280.404\end{aligned}$$

For the house with an attached garage, a 1 is substituted for x_4 in the regression equation. The estimated heating cost is $357.836, found by:

$$\begin{aligned}\hat{y} &= 393.666 - 3.963x_1 - 11.334x_2 + 77.432x_4 \\ &= 393.666 - 3.963(20) - 11.334(3) + 77.432(1) = 357.836\end{aligned}$$

The difference between the estimated heating costs is $77.432 ($357.836 − $280.404). Hence, we can expect the cost to heat a house with an attached garage to be $77.432 more than the cost for an equivalent house without a garage.

We have shown the difference between the two types of homes to be $77.432, but is the difference significant? We conduct the following test of hypothesis.

$$H_0: \beta_4 = 0$$
$$H_1: \beta_4 \neq 0$$

The information necessary to answer this question is in the output at the bottom of the previous page. The regression coefficient for the independent variable garage is $77.432, and the standard deviation of the sampling distribution is 22.783. We identify this as the fourth independent variable, so we use a subscript of 4. (Remember we dropped age of the furnace, the third independent variable.) Finally, we insert these values in formula (14–6).

$$t = \frac{b_4 - 0}{s_{b_4}} = \frac{77.432 - 0}{22.783} = 3.399$$

There are three independent variables in the analysis, so there are $n - (k + 1) = 20 - (3 + 1) = 16$ degrees of freedom. The critical value from Appendix B.5 is 2.120. The decision rule, using a two-tailed test and the .05 significance level, is to reject H_0 if the computed t is to the left of −2.120 or to the right of 2.120. Because the computed value of 3.399 is to the right of 2.120, the null hypothesis is rejected. We conclude that the regression coefficient is not zero. The independent variable garage should be included in the analysis.

Using the p-value approach, the computed t value of 3.399 has a p-value of 0.004. This value is less than the .05 significance level. Therefore, we reject the null hypothesis. We conclude that the regression coefficient is not zero and the independent variable garage should be included in the analysis.

Is it possible to use a qualitative variable with more than two possible outcomes? Yes, but the coding scheme becomes more complex and will require a series of dummy variables. To explain, suppose a company is studying its sales as they relate to advertising expense by quarter for the last 5 years. Let sales be the dependent variable and advertising expense be the first independent variable, x_1. To include the qualitative information regarding the quarter, we use three additional independent variables. For the variable x_2, the five observations referring to the first quarter of each of the 5 years are coded 1 and the other quarters 0. Similarly, for x_3 the five observations referring to the second quarter are coded 1 and the other quarters 0. For x_4, the five observations referring to the third quarter are coded 1 and the other quarters 0. An observation that does not refer to any of the first three quarters must refer to the fourth quarter, so a distinct independent variable referring to this quarter is not necessary.

SELF-REVIEW 14–4

A study by the American Realtors Association investigated the relationship between the commissions earned by sales associates last year and the number of months since the associates earned their real estate licenses. Also of interest in the study is the gender of the sales associate. Below is a portion of the regression output. The dependent variable is commissions, which is reported in $000, and the independent variables are months since the license was earned and gender (female = 1 and male = 0).

Regression Analysis

Regression Statistics	
Multiple R	0.801
R Square	0.642
Adjusted R Square	0.600
Standard Error	3.219
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*p-value*
Regression	2	315.9291	157.9645	15.2468	0.0002
Residual	17	176.1284	10.36049		
Total	19	492.0575			

	Coefficients	*Standard Error*	*t Stat*	*p-value*
Intercept	15.7625	3.0782	5.121	.0001
Months	0.4415	0.0839	5.262	.0001
Gender	3.8598	1.4724	2.621	.0179

(a) Write out the regression equation. How much commission would you expect a female agent to make who earned her license 30 months ago?
(b) Do the female agents on the average make more or less than the male agents? How much more?
(c) Conduct a test of hypothesis to determine if the independent variable gender should be included in the analysis. Use the .05 significance level. What is your conclusion?

LO14-6
Include and interpret an interaction effect in a multiple regression analysis.

REGRESSION MODELS WITH INTERACTION

In Chapter 12, we discussed interaction among independent variables. To explain, suppose we are studying weight loss and assume, as the current literature suggests, that diet and exercise are related. So the dependent variable is amount of change in weight and the independent variables are diet (yes or no) and exercise (none, moderate, significant). We are interested in whether there is interaction among the independent variables. That is, if those studied maintain their diet and exercise significantly, will that increase the mean amount of weight lost? Is total weight loss more than the sum of the loss due to the diet effect and the loss due to the exercise effect?

We can expand on this idea. Instead of having two nominal-scale variables, diet and exercise, we can examine the effect (interaction) of several ratio-scale variables. For example, suppose we want to study the effect of room temperature (68, 72, 76, or 80 degrees Fahrenheit) and noise level (60, 70, or 80 decibels) on the number of units produced. To put it another way, does the combination of noise level in the room and the temperature of the room have an effect on the productivity of the workers? Would workers produce more units in a quiet, cool room compared to a hot, noisy room?

In regression analysis, interaction is examined as a separate independent variable. An interaction prediction variable can be developed by multiplying the data values of one independent variable by the values of another independent variable, thereby creating a new independent variable. A two-variable model that includes an interaction term is:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

The term X_1X_2 is the *interaction term.* We create this variable by multiplying the values of X_1 and X_2 to create a third independent variable. We then develop a regression equation using the three independent variables and test the significance of the third independent variable using the individual test for independent variables, described earlier in the chapter. An example will illustrate the details.

EXAMPLE

Refer to the heating cost example and the data in Table 14–1. Is there an interaction between the outside temperature and the amount of insulation? If both variables are increased, is the effect on heating cost greater than the sum of savings from warmer temperature and the savings from increased insulation separately?

SOLUTION

The information from Table 14–1 for the independent variables temperature and insulation is repeated below. We create the interaction variable by multiplying the value of temperature by the value insulation for each observation in the data set. For the first sampled home, the value temperature is 35 degrees and insulation is

3 inches so the value of the interaction variable is $35 \times 3 = 105$. The values of the other interaction products are found in a similar fashion.

	A	B	C	D	E	F	G	H	I	J	K
1	Cost	Temp	Insul	Temp X Insul		SUMMARY OUTPUT					
2	250	35	3	105							
3	360	29	4	116		*Regression Statistics*					
4	165	36	7	252		Multiple R	0.893				
5	43	60	6	360		R Square	0.798				
6	92	65	5	325		Adjusted R Square	0.760				
7	200	30	5	150		Standard Error	51.846				
8	355	10	6	60		Observations	20				
9	290	7	10	70							
10	230	21	9	189		ANOVA					
11	120	55	2	110			*df*	*SS*	*MS*	*F*	*Significance F*
12	73	54	12	648		Regression	3	169908.452	56636.151	21.070	0.000
13	205	48	5	240		Residual	16	43007.298	2687.956		
14	400	20	5	100		Total	19	212915.750			
15	320	39	4	156							
16	72	60	8	480			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	272	20	5	100		Intercept	598.070	92.265	6.482	0.000	
18	94	58	7	406		Temp	-7.811	2.124	-3.678	0.002	
19	190	40	8	320		Insul	-30.161	12.621	-2.390	0.030	
20	235	27	9	243		Temp X Insul	0.385	0.291	1.324	0.204	
21	139	30	7	210							

We find the multiple regression using temperature, insulation, and the interaction of temperature and insulation as independent variables. The regression equation is reported below.

$$\hat{y} = 598.070 - 7.811x_1 - 30.161x_2 + 0.385x_1x_2$$

The question we wish to answer is whether the interaction variable is significant. Note we use the subscript, 1×2, to indicate the coefficient of the interaction of variables 1 and 2. We will use the .05 significance level. In terms of a hypothesis:

$$H_0: \beta_{1\times 2} = 0$$

$$H_1: \beta_{1\times 2} \neq 0$$

There are $n - (k + 1) = 20 - (3 + 1) = 16$ degrees of freedom. Using the .05 significance level and a two-tailed test, the critical values of t are -2.120 and 2.120. We reject the null hypothesis if t is less than -2.120 or t is greater than 2.120. From the output, $b_{1\times 2} = 0.385$ and $s_{b_{1x2}} = 0.291$. To find the value of t, we use formula (14–6).

$$t = \frac{b_{1x2} - 0}{s_{b_{1x2}}} = \frac{0.385 - 0}{0.291} = 1.324$$

Because the computed value of 1.324 is less than the critical value of 2.120, we do not reject the null hypothesis. In addition, the p-value of .204 exceeds .05. We conclude that there is not a significant interaction effect of temperature and insulation on home heating costs.

There are other situations that can occur when studying interaction among independent variables.

1. It is possible to have a three-way interaction among the independent variables. In our heating example, we might have considered the three-way interaction between temperature, insulation, and age of the furnace.
2. It is possible to have an interaction where one of the independent variables is nominal scale. In our heating cost example, we could have studied the interaction between temperature and garage.

Studying all possible interactions can become very complex. However, careful consideration to possible interactions among independent variables can often provide useful insight into the regression models.

LO14-7
Apply stepwise regression to develop a multiple regression model.

STEPWISE REGRESSION

In our heating cost example (see sample information in Table 14–1), we considered three independent variables: the mean outside temperature, the amount of insulation in the home, and the age of the furnace. To obtain the equation, we first ran a global or "all at once" test to determine if any of the regression coefficients were significant. When we found at least one to be significant, we tested the regression coefficients individually to determine which were important. We kept the independent variables that had significant regression coefficients and left the others out. By retaining the independent variables with significant coefficients, we found the regression equation that used the fewest independent variables. This made the regression equation easier to interpret. Then we considered the qualitative variable, garage, and found that it was significantly related to heating cost. The variable, garage, was added to the equation.

Deciding the set of independent variables to include in a multiple regression equation can be accomplished using a a technique called **stepwise regression.** This technique efficiently builds an equation that only includes independent variables with significant regression coefficients.

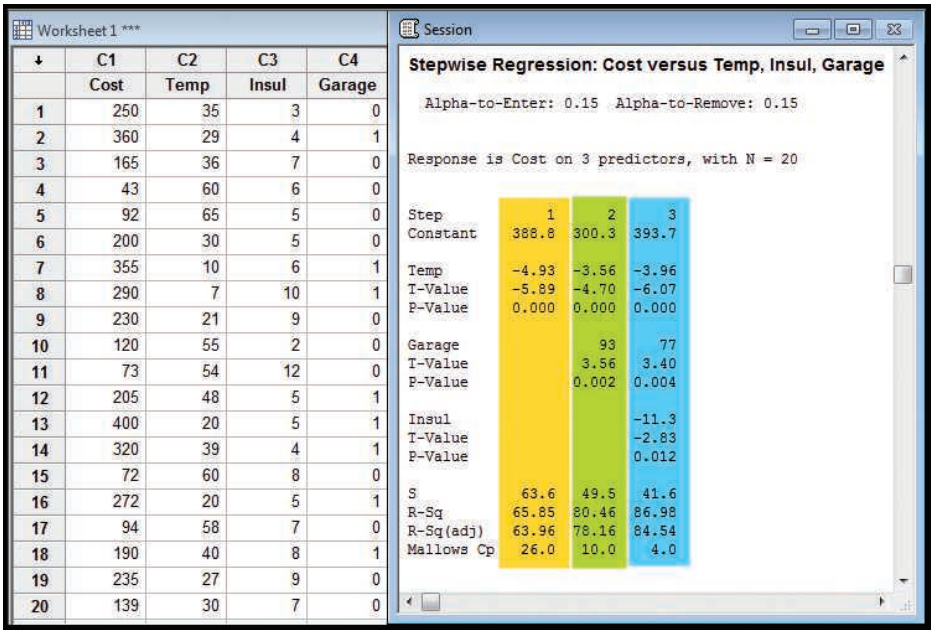
STEPWISE REGRESSION A step-by-step method to determine a regression equation that begins with a single independent variable and adds or deletes independent variables one by one. Only independent variables with nonzero regression coefficients are included in the regression equation.

In the stepwise method, we develop a sequence of equations. The first equation contains only one independent variable. However, this independent variable is the one from the set of proposed independent variables that explains the most variation in the dependent variable. Stated differently, if we compute all the simple correlations between each independent variable and the dependent variable, the stepwise method first selects the independent variable with the strongest correlation with the dependent variable.

Next, the stepwise method looks at the remaining independent variables and then selects the one that will explain the largest percentage of the variation yet unexplained. We continue this process until all the independent variables with significant regression coefficients are included in the regression equation. The advantages to the stepwise method are:

1. Only independent variables with significant regression coefficients are entered into the equation.
2. The steps involved in building the regression equation are clear.
3. It is efficient in finding the regression equation with only significant regression coefficients.
4. The changes in the multiple standard error of estimate and the coefficient of determination are shown.

Stepwise regression procedures are included in many statistical software packages. For example, Minitab's stepwise regression analysis for the home heating cost problem follows. Note that the final equation, which is reported in the column labeled 3, includes the independent variables temperature, garage, and insulation. These are the same independent variables that were included in our equation using the global test and the test for individual independent variables. The independent variable age, indicating the furnace's age, is not included because it is not a significant predictor of cost.



Worksheet 1 ***

↓	C1	C2	C3	C4
	Cost	Temp	Insul	Garage
1	250	35	3	0
2	360	29	4	1
3	165	36	7	0
4	43	60	6	0
5	92	65	5	0
6	200	30	5	0
7	355	10	6	1
8	290	7	10	1
9	230	21	9	0
10	120	55	2	0
11	73	54	12	0
12	205	48	5	1
13	400	20	5	1
14	320	39	4	1
15	72	60	8	0
16	272	20	5	1
17	94	58	7	0
18	190	40	8	1
19	235	27	9	0
20	139	30	7	0

Session

Stepwise Regression: Cost versus Temp, Insul, Garage

```
 Alpha-to-Enter: 0.15  Alpha-to-Remove: 0.15


Response is Cost on 3 predictors, with N = 20


Step             1       2       3
Constant     388.8   300.3   393.7

Temp         -4.93   -3.56   -3.96
T-Value      -5.89   -4.70   -6.07
P-Value      0.000   0.000   0.000

Garage                  93      77
T-Value               3.56    3.40
P-Value              0.002   0.004

Insul                        -11.3
T-Value                      -2.83
P-Value                      0.012

S             63.6    49.5    41.6
R-Sq         65.85   80.46   86.98
R-Sq(adj)    63.96   78.16   84.54
Mallows Cp    26.0    10.0     4.0
```

Reviewing the steps and interpreting output:

1. The stepwise procedure selects the independent variable temperature first. This variable explains more of the variation in heating cost than any of the other three proposed independent variables. Temperature explains 65.85% of the variation in heating cost. The regression equation is:

$$\hat{y} = 388.8 - 4.93x_1$$

   There is an inverse relationship between heating cost and temperature. For each degree the temperature increases, heating cost is reduced by \$4.93.

2. The next independent variable to enter the regression equation is garage. When this variable is added to the regression equation, the coefficient of determination is increased from 65.85% to 80.46%. That is, by adding garage as an independent variable, we increase the coefficient of determination by 14.61 percentage points. The regression equation after step 2 is:

$$\hat{y} = 300.3 - 3.56x_1 + 93.0x_2$$

   Usually the regression coefficients will change from one step to the next. In this case, the coefficient for temperature retained its negative sign, but it changed from −4.93 to −3.56. This change is reflective of the added influence of the independent variable garage. Why did the stepwise method select the independent variable garage instead of either insulation or age? The increase in R^2, the coefficient of determination, is larger if garage is included rather than either of the other two variables.

3. At this point, there are two unused variables remaining, insulation and age. Notice on the third step the procedure selects insulation and then stops. This indicates the variable insulation explains more of the remaining variation in heating cost than the age variable does. After the third step, the regression equation is:

$$\hat{y} = 393.7 - 3.96x_1 + 77.0x_2 - 11.3x_3$$

   At this point, 86.98% of the variation in heating cost is explained by the three independent variables temperature, garage, and insulation. This is the same R^2 value and regression equation we found on page 512 except for rounding differences.

4. Here, the stepwise procedure stops. This means the independent variable age does not add significantly to the coefficient of determination.

The stepwise method developed the same regression equation, selected the same independent variables, and found the same coefficient of determination as the global and individual tests described earlier in the chapter. The advantage to the stepwise method is that it is more direct than using a combination of the global and individual procedures.

Other methods of variable selection are available. The stepwise method is also called the **forward selection method** because we begin with no independent variables and add one independent variable to the regression equation at each iteration. There is also the **backward elimination method,** which begins with the entire set of variables and eliminates one independent variable at each iteration.

The methods described so far look at one variable at a time and decide whether to include or eliminate that variable. Another approach is the **best-subset regression.** With this method, we look at the best model using one independent variable, the best model using two independent variables, the best model with three, and so on. The criterion is to find the model with the largest R^2 value, regardless of the number of independent variables. Also, each independent variable does not necessarily have a nonzero regression coefficient. Since each independent variable could either be included or not included, there are $2^k - 1$ possible models, where k refers to the number of independent variables. In our heating cost example, we considered four independent variables so there are 15 possible regression models, found by $2^4 - 1 = 16 - 1 = 15$. We would examine all regression models using one independent variable, all combinations using two variables, all combinations using three independent variables, and the possibility of using all four independent variables. The advantages to the best-subset method is it may examine combinations of independent variables not considered in the stepwise method. The process is available in Minitab and MegaStat.

EXERCISES

9. FILE The manager of High Point Sofa and Chair, a large furniture manufacturer located in North Carolina, is studying the job performance ratings of a sample of 15 electrical repairmen employed by the company. An aptitude test is required by the human resources department to become an electrical repairman. The manager was able to get the score for each repairman in the sample. In addition, he determined which of the repairmen were union members (code = 1) and which were not (code = 0). The sample information is reported below.

Worker	Job Performance Score	Aptitude Test Score	Union Membership
Abbott	58	5	0
Anderson	53	4	0
Bender	33	10	0
Bush	97	10	0
Center	36	2	0
Coombs	83	7	0
Eckstine	67	6	0
Gloss	84	9	0
Herd	98	9	1
Householder	45	2	1
Lori	97	8	1
Lindstrom	90	6	1
Mason	96	7	1
Pierse	66	3	1
Rohde	82	6	1

a. Use a statistical software package to develop a multiple regression equation using the job performance score as the dependent variable and aptitude test score and union membership as independent variables.

b. Comment on the regression equation. Be sure to include the coefficient of determination and the effect of union membership. Are these two variables effective in explaining the variation in job performance?

c. Conduct a test of hypothesis to determine if union membership should be included as an independent variable.

d. Repeat the analysis considering possible interaction terms.

10. **FILE** Cincinnati Paint Company sells quality brands of paints through hardware stores throughout the United States. The company maintains a large sales force who call on existing customers and look for new business. The national sales manager is investigating the relationship between the number of sales calls made and the miles driven by the sales representative. Also, do the sales representatives who drive the most miles and make the most calls necessarily earn the most in sales commissions? To investigate, the vice president of sales selected a sample of 25 sales representatives and determined:

- The amount earned in commissions last month (y)
- The number of miles driven last month (x_1)
- The number of sales calls made last month (x_2)

The information is reported below.

Commissions ($000)	Calls	Driven
22	139	2,371
13	132	2,226
33	144	2,731
⋮	⋮	⋮
25	127	2,671
43	154	2,988
34	147	2,829

Develop a regression equation including an interaction term. Is there a significant interaction between the number of sales calls and the miles driven?

11. **FILE** An art collector is studying the relationship between the selling price of a painting and two independent variables. The two independent variables are the number of bidders at the particular auction and the age of the painting, in years. A sample of 25 paintings revealed the following sample information.

Painting	Auction Price	Bidders	Age
1	3,470	10	67
2	3,500	8	56
3	3,700	7	73
⋮	⋮	⋮	⋮
23	4,660	5	94
24	4,710	3	88
25	4,880	1	84

a. Develop a multiple regression equation using the independent variables number of bidders and age of painting to estimate the dependent variable auction price. Discuss the equation. Does it surprise you that there is an inverse relationship between the number of bidders and the price of the painting?

b. Create an interaction variable and include it in the regression equation. Explain the meaning of the interaction. Is this variable significant?

c. Use the stepwise method and the independent variables for the number of bidders, the age of the painting, and the interaction between the number of bidders and the age of the painting. Which variables would you select?

12. FILE A real estate developer wishes to study the relationship between the size of home a client will purchase (in square feet) and other variables. Possible independent variables include the family income, family size, whether there is a senior adult parent living with the family (1 for yes, 0 for no), and the total years of education beyond high school for the husband and wife. The sample information is reported below.

Family	Square Feet	Income (000s)	Family Size	Senior Parent	Education
1	2,240	60.8	2	0	4
2	2,380	68.4	2	1	6
3	3,640	104.5	3	0	7
4	3,360	89.3	4	1	0
5	3,080	72.2	4	0	2
6	2,940	114	3	1	10
7	4,480	125.4	6	0	6
8	2,520	83.6	3	0	8
9	4,200	133	5	0	2
10	2,800	95	3	0	6

Develop an appropriate multiple regression equation. Which independent variables would you include in the final regression equation? Use the stepwise method.

LO14-8
Apply multiple regression techniques to develop a linear model.

REVIEW OF MULTIPLE REGRESSION

We described many topics involving multiple regression in this chapter. In this section of the chapter, we focus on a single example with a solution that reviews the procedure and guides your application of multiple regression analysis.

EXAMPLE

The Bank of New England is a large financial institution serving the New England states as well as New York and New Jersey. The mortgage department of the Bank of New England is studying data from recent loans. Of particular interest is how such factors as the value of the home being purchased ($000), education level of the head of the household (number of years, beginning with first grade), age of the head of the household, current monthly mortgage payment (in dollars), and gender of the head of the household (male = 1, female = 0) relate to the family income. The mortgage department would like to know whether these variables are effective predictors of family income.

SOLUTION

FILE Consider a random sample of 25 loan applications submitted to the Bank of New England last month. A portion of the sample information is shown in Table 14–3. The entire data set is available at the website (www.mhhe.com/Lind17e) and is identified as Bank of New England.

TABLE 14–3 Information on Sample of 25 Loans by the Bank of New England

Loan	Income ($000)	Value ($000)	Education	Age	Mortgage	Gender
1	100.7	190	14	53	230	1
2	99.0	121	15	49	370	1
3	102.0	161	14	44	397	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮
23	102.3	163	14	46	142	1
24	100.2	150	15	50	343	0
25	96.3	139	14	45	373	0

We begin by calculating the correlation matrix shown below. It shows the relationship between each of the independent variables and the dependent variable. It helps to identify the independent variables that are more closely related to the dependent variable (family income). The correlation matrix also reveals the independent variables that are highly correlated and possibly redundant.

	Income	Value	Education	Age	Mortgage	Gender
Income	1					
Value	0.720	1				
Education	0.188	–0.144	1			
Age	0.243	0.220	0.621	1		
Mortgage	0.116	0.358	–0.210	–0.038	1	
Gender	0.486	0.184	0.062	0.156	–0.129	1

What can we learn from this correlation matrix?

1. The first column shows the correlations between each of the independent variables and the dependent variable family income. Observe that each of the independent variables is positively correlated with family income. The value of the home has the strongest correlation with family income. The level of education of the person applying for the loan and the current mortgage payment have a weak correlation with family income. These two variables are candidates to be dropped from the regression equation.
2. All possible correlations among the independent variables are identified with the green background. Our standard is to look for correlations that exceed an absolute value of .700. None of the independent variables are strongly correlated with each other. This indicates that multicollinearity is not likely.

Next, we compute the multiple regression equation using all the independent variables. The software output follows.

	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	*Regression Statistics*					
4	Multiple R	0.866				
5	R Square	0.750				
6	Adjusted R Square	0.684				
7	Standard Error	1.478				
8	Observations	25				
9						
10	ANOVA					
11		*df*	*SS*	*MS*	*F*	*P-value*
12	Regression	5	124.322	24.864	11.385	0.000
13	Residual	19	41.494	2.184		
14	Total	24	165.815			
15						
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	Intercept	70.606	7.464	9.459	0.000	
18	Value ($000)	0.072	0.012	5.769	0.000	
19	Education	1.624	0.603	2.693	0.014	
20	Age	-0.122	0.078	-1.566	0.134	
21	Mortgage	-0.001	0.003	-0.319	0.753	
22	Gender	1.807	0.623	2.901	0.009	

The coefficients of determination, that is, both R^2 and adjusted R^2, are reported at the top of the summary output and highlighted in yellow. The R^2 value is 75.0%, so the five independent variables account for three-quarters of the variation in family income. The adjusted R^2 measures the strength of the relationship between the set of independent variables and family income and also accounts for the number of variables in the regression equation. The adjusted R^2 indicates that the five variables account for 68.4% of the variance of family income. Both of these suggest that the proposed independent variables are useful in predicting family income.

The output also includes the regression equation.

$$\hat{y} = 70.606 + 0.072(Value) + 1.624(Education) - 0.122(Age) \\ - 0.001(Mortgage) + 1.807(Gender)$$

Be careful in this interpretation. Both income and the value of the home are in thousands of dollars. Here is a summary:

1. An increase of $1,000 in the value of the home suggests an increase of $72 in family income. An increase of 1 year of education increases income by $1,624, another year older reduces income by $122, and an increase of $1,000 in the mortgage reduces income by $1.
2. If a male is head of the household, the value of family income will increase by $1,807. Remember that "female" was coded 0 and "male" was coded 1, so a male head of household is positively related to home value.
3. The age of the head of household and monthly mortgage payment are inversely related to family income. This is true because the sign of the regression coefficient is negative.

Next we conduct the global hypothesis test. Here we check to see if any of the regression coefficients are different from 0. We use the .05 significance level.

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$
$$H_1: \text{Not all the } \beta\text{'s are } 0$$

The p-value from the table (cell F12) is 0.000. Because the p-value is less than the significance level, we reject the null hypothesis and conclude that at least one of the regression coefficients is not equal to zero.

Next we evaluate the individual regression coefficients. The p-values to test each regression coefficient are reported in cells E18 through E22 in the software output on the previous page. The null hypothesis and the alternate hypothesis are:

$$H_0: \beta_i = 0$$
$$H_1: \beta_i \neq 0$$

The subscript i represents any particular independent variable. Again using .05 significance levels, the p-values for the regression coefficients for home value, years of education, and gender are all less than .05. We conclude that these regression coefficients are not equal to zero and are significant predictors of family income. The p-value for age and mortgage amount are greater than the significance level of .05, so we do not reject the null hypotheses for these variables. The regression coefficients are not different from zero and are not related to family income.

Based on the results of testing each of the regression coefficients, we conclude that the variables age and mortgage amount are not effective predictors of family income. Thus, they should be removed from the multiple regression equation. Remember that we must remove one independent variable at a time and redo the analysis to evaluate the overall effect of removing the variable. Our strategy is to remove the variable with the smallest t-statistic or the largest p-value. This variable is mortgage amount. The result of the regression analysis without the mortgage variable follows.

	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	*Regression Statistics*					
4	Multiple R	0.865				
5	R Square	0.748				
6	Adjusted R Square	0.698				
7	Standard Error	1.444				
8	Observations	25				
9						
10	ANOVA					
11		*df*	*SS*	*MS*	*F*	*P-value*
12	Regression	4	124.099	31.025	14.874	0.000
13	Residual	20	41.716	2.086		
14	Total	24	165.815			
15						
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	Intercept	70.159	7.165	9.791	0.000	
18	Value ($000)	0.070	0.011	6.173	0.000	
19	Education	1.647	0.585	2.813	0.011	
20	Age	-0.122	0.076	-1.602	0.125	
21	Gender	1.846	0.596	3.096	0.006	

Observe that the R^2 and adjusted R^2 change very little without the mortgage variable. Also observe that the *p*-value associated with age is greater than the .05 significance level. So next we remove the age variable and redo the analysis. The regression output with the variables age and mortgage amount removed follows.

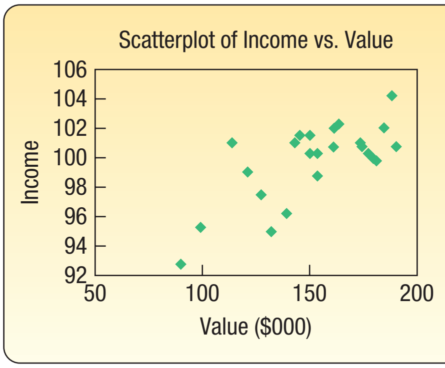
	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	*Regression Statistics*					
4	Multiple R	0.846				
5	R Square	0.716				
6	Adjusted R Square	0.676				
7	Standard Error	1.497				
8	Observations	25				
9						
10	ANOVA					
11		*df*	*SS*	*MS*	*F*	*P-value*
12	Regression	3	118.743	39.581	17.658	0.000
13	Residual	21	47.072	2.242		
14	Total	24	165.815			
15						
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17	Intercept	74.527	6.870	10.849	0.000	
18	Value ($000)	0.063	0.011	5.803	0.000	
19	Education	1.016	0.449	2.262	0.034	
20	Gender	1.770	0.616	2.872	0.009	

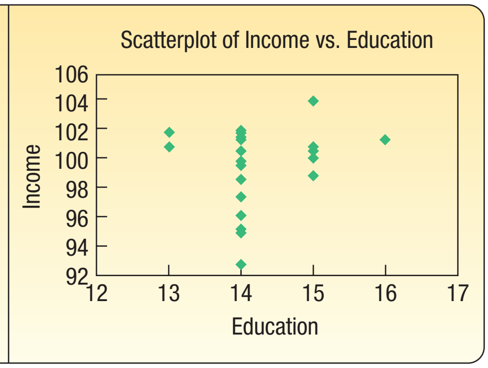
From this output, we conclude:

1. The R^2 and adjusted R^2 values have declined but only slightly. Using all five independent variables, the R^2 value was .750. With the two nonsignificant variables removed, the R^2 and adjusted R^2 values are .716 and .676, respectively. We prefer the equation with the fewer number of independent variables. It is easier to interpret.
2. In ANOVA, we observe that the *p*-value is less than .05. Hence, at least one of the regression coefficients is not equal to zero.

3. Reviewing the significance of the individual coefficients, the p-values associated with each of the remaining independent variables are less than .05. We conclude that all the regression coefficients are different from zero. Each independent variable is a useful predictor of family income.

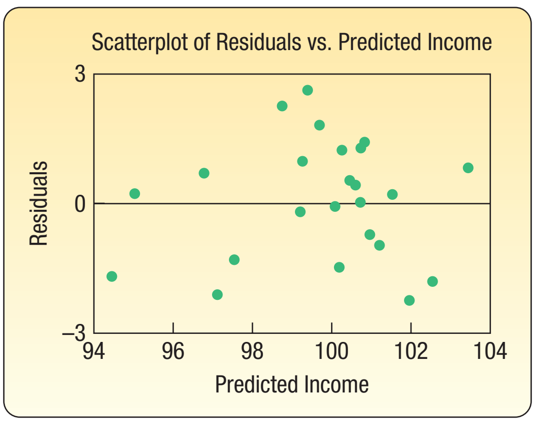
Our final step is to examine the regression assumptions (Evaluating the Assumptions of Multiple Regression section on page 506) with our regression model. The first assumption is that there is a linear relationship between each independent variable and the dependent variable. It is not necessary to review the dummy variable Gender because there are only two possible outcomes. Below are the scatter plots of family income versus home value and family income versus years of education.





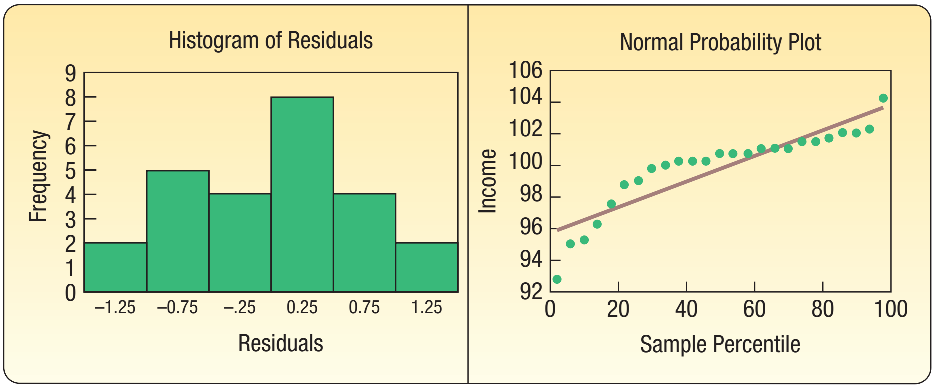
The scatter plot of income versus home value shows a general increasing trend. As the home value increases, so does family income. The points appear to be linear. That is, there is no observable nonlinear pattern in the data. The scatter plot on the right, of income versus years of education, shows that the data are measured to the nearest year. The measurement is to the nearest year and is a discrete variable. Given the measurement method, it is difficult to determine if the relationship is linear or not.

A plot of the residuals is also useful to evaluate the overall assumption of linearity. Recall that a residual is $(y - \hat{y})$, the difference between the actual value of the dependent variable (y) and the predicted value of the dependent variable ($\hat{y}$). Assuming a linear relationship, the distribution of the residuals should show about an equal proportion of negative residuals (points above the line) and positive residuals (points below the line) centered on zero. There should be no observable pattern to the plots. The graph follows.



There is no discernable pattern to the plot, so we conclude that the linearity assumption is reasonable.

If the linearity assumption is valid, then the distribution of residuals should follow the normal probability distribution with a mean of zero. To evaluate this assumption, we will use a histogram and a normal probability plot.



In general, the histogram on the left shows the major characteristics of a normal distribution, that is, a majority of observations in the middle and centered on the mean of zero, with lower frequencies in the tails of the distribution. The normal probability plot on the right is based on a cumulative normal probability distribution. The line shows the standardized cumulative normal distribution. The green dots show the cumulative distribution of the residuals. To confirm the normal distribution of the residuals, the green dots should be close to the line. This is true for most of the plot. However, we would note that there are departures and even perhaps a nonlinear pattern in the residuals in the lower part of the graph. As before, we are looking for serious departures from linearity and these are not indicated in these graphs.

The final assumption refers to multicollinearity. This means that the independent variables should not be highly correlated. We suggested a rule of thumb that multicollinearity would be a concern if the correlations among independent variables were close to 0.7 or −0.7. There are no violations of this guideline.

There is a statistic that is used to more precisely evaluate multicollinearity, the variance inflation factor (*VIF*). To calculate the *VIF*s, we need to do a regression analysis for each independent variable as a function of the other independent variables. From each of these regression analyses, we need the R^2 to compute the *VIF* using formula (14–7). The following table shows the R^2 for each regression analysis and the computed *VIF*. If the *VIF*s are less than 10, then multicollinearity is not a concern. In this case, the *VIF*s are all less than 10, so multicollinearity among the independent variables is not a concern.

Dependent Variable	Independent Variables	*R*-square	*VIF*
Value	Education and Gender	0.058	1.062
Education	Gender and Value	0.029	1.030
Gender	Value and Education	0.042	1.044

To summarize, the multiple regression equation is

$$\hat{y} = 74.527 + 0.063(Value) + 1.016(Education) + 1.770(Gender)$$

This equation explains 71.6% of the variation in family income. There are no major departures from the multiple regression assumptions of linearity, normally distributed residuals, and multicollinearity.

CHAPTER SUMMARY

I. The general form of a multiple regression equation is:

$$\hat{y} = a + b_1x_1 + b_2x_2 + \ldots + b_kx_k \quad \textbf{(14–1)}$$

where a is the Y-intercept when all x's are zero, b_j refers to the sample regression coefficients, and x_j refers to the value of the various independent variables.

A. There can be any number of independent variables.

B. The least squares criterion is used to develop the regression equation.

C. A statistical software package is needed to perform the calculations.

II. An ANOVA table summarizes the multiple regression analysis.

A. It reports the total amount of the variation in the dependent variable and divides this variation into that explained by the set of independent variables and that not explained.

B. It reports the degrees of freedom associated with the independent variables, the error variation, and the total variation.

III. There are two measures of the effectiveness of the regression equation.

A. The multiple standard error of estimate is similar to the standard deviation.

1. It is measured in the same units as the dependent variable.
2. It is based on squared deviations between the observed and predicted values of the dependent variable.
3. It ranges from 0 to plus infinity.
4. It is calculated from the following equation.

$$s_{y.123\ldots k} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - (k + 1)}} \quad \textbf{(14–2)}$$

B. The coefficient of multiple determination reports the percent of the variation in the dependent variable explained by the variation in the set of independent variables.

1. It may range from 0 to 1.
2. It is also based on squared deviations from the regression equation.
3. It is found by the following equation.

$$R^2 = \frac{\text{SSR}}{\text{SS total}} \quad \textbf{(14–3)}$$

4. When the number of independent variables is large, we adjust the coefficient of determination for the degrees of freedom as follows.

$$R^2_{\text{adj}} = 1 - \frac{\dfrac{\text{SSE}}{n - (k + 1)}}{\dfrac{\text{SS total}}{n - 1}} \quad \textbf{(14–4)}$$

IV. A global test is used to investigate whether any of the independent variables have a regression coefficient that differs significantly from zero.

A. The null hypothesis is: All the regression coefficients are zero.

B. The alternate hypothesis is: At least one regression coefficient is not zero.

C. The test statistic is the F distribution with k (the number of independent variables) degrees of freedom in the numerator and $n - (k + 1)$ degrees of freedom in the denominator, where n is the sample size.

D. The formula to calculate the value of the test statistic for the global test is:

$$F = \frac{\text{SSR}/k}{\text{SSE}/[n - (k + 1)]} \quad \textbf{(14–5)}$$

V. The test for individual variables determines which independent variables have regression coefficients that differ significantly from zero.

A. The variables that have zero regression coefficients are usually dropped from the analysis.

B. The test statistic is the t distribution with $n - (k + 1)$ degrees of freedom.

C. The formula to calculate the value of the test statistic for the individual test is:

$$t = \frac{b_i - 0}{s_{b_i}} \qquad \textbf{(14–6)}$$

VI. There are five assumptions to use multiple regression analysis.

A. The relationship between the dependent variable and the set of independent variables must be linear.

1. To verify this assumption, develop a scatter diagram and plot the residuals on the vertical axis and the fitted values on the horizontal axis.
2. If the plots appear random, we conclude the relationship is linear.

B. The variation is the same for both large and small values of $\hat{y}$.

1. Homoscedasticity means the variation is the same for all fitted values of the dependent variable.
2. This condition is checked by developing a scatter diagram with the residuals on the vertical axis and the fitted values on the horizontal axis.
3. If there is no pattern to the plots—that is, they appear random—the residuals meet the homoscedasticity requirement.

C. The residuals follow the normal probability distribution.

1. This condition is checked by developing a histogram of the residuals or a normal probability plot.
2. The mean of the distribution of the residuals is 0.

D. The independent variables are not correlated.

1. A correlation matrix will show all possible correlations among independent variables. Signs of trouble are correlations larger than 0.70 or less than −0.70.
2. Signs of correlated independent variables include when an important predictor variable is found insignificant, when an obvious reversal occurs in signs in one or more of the independent variables, or when a variable is removed from the solution, there is a large change in the regression coefficients.
3. The variance inflation factor is used to identify correlated independent variables.

$$VIF = \frac{1}{1 - R_j^2} \qquad \textbf{(14–7)}$$

E. Each residual is independent of other residuals.

1. Autocorrelation occurs when successive residuals are correlated.
2. When autocorrelation exists, the value of the standard error will be biased and will return poor results for tests of hypothesis regarding the regression coefficients.

VII. Several techniques help build a regression model.

A. A dummy or qualitative independent variable can assume one of two possible outcomes.

1. A value of 1 is assigned to one outcome and 0 to the other.
2. Use formula (14–6) to determine if the dummy variable should remain in the equation.

B. Interaction is the case in which one independent variable (such as x_2) affects the relationship with another independent variable (x_1) and the dependent variable (y).

C. Stepwise regression is a step-by-step process to find the regression equation.

1. Only independent variables with nonzero regression coefficients enter the equation.
2. Independent variables are added one at a time to the regression equation.

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
b_1	Regression coefficient for the first independent variable	*b sub 1*
b_k	Regression coefficient for any independent variable	*b sub k*
$s_{y.123...k}$	Multiple standard error of estimate	*s sub y dot 1, 2, 3 . . . k*

CHAPTER EXERCISES

13. A multiple regression analysis yields the following partial results.

Source	Sum of Squares	*df*
Regression	750	4
Error	500	35

a. What is the total sample size?
b. How many independent variables are being considered?
c. Compute the coefficient of determination.
d. Compute the standard error of estimate.
e. Test the hypothesis that at least one of the regression coefficients is not equal to zero. Let $\alpha = .05$.

14. In a multiple regression analysis, two independent variables are considered, and the sample size is 25. The regression coefficients and the standard errors are as follows.

$$b_1 = 2.676 \qquad s_{b_1} = 0.56$$
$$b_2 = -0.880 \qquad s_{b_2} = 0.71$$

Conduct a test of hypothesis to determine whether either independent variable has a coefficient equal to zero. Would you consider deleting either variable from the regression equation? Use the .05 significance level.

15. The following output was obtained from a multiple regression analysis.

Analysis of Variance			
Source	DF	SS	MS
Regression	5	100	20
Residual Error	20	40	2
Total	25	140	
Predictor	Coefficient	SE Coefficient	t
Constant	3.00	1.50	2.00
x_1	4.00	3.00	1.33
x_2	3.00	0.20	15.00
x_3	0.20	0.05	4.00
x_4	-2.50	1.00	-2.50
x_5	3.00	4.00	0.75

a. What is the sample size?
b. Compute the value of R^2.
c. Compute the multiple standard error of estimate.
d. Conduct a global test of hypothesis to determine whether any of the regression coefficients are significant. Use the .05 significance level.
e. Test the regression coefficients individually. Would you consider omitting any variable(s)? If so, which one(s)? Use the .05 significance level.

16. In a multiple regression analysis, $k = 5$ and $n = 20$, the MSE value is 5.10, and SS total is 519.68. At the .05 significance level, can we conclude that any of the regression coefficients are not equal to 0?

17. The district manager of Jasons, a large discount electronics chain, is investigating why certain stores in her region are performing better than others. She believes that three factors are related to total sales: the number of competitors in the region, the population in the surrounding area, and the amount spent on advertising. From her district, consisting of several hundred stores, she selects a random sample of 30 stores. For each store, she gathered the following information.

y = total sales last year (in \$ thousands)
x_1 = number of competitors in the region
x_2 = population of the region (in millions)
x_3 = advertising expense (in \$ thousands)

The results of a multiple regression analysis, using Minitab, follow.

Analysis of Variance			
Source	DF	SS	MS
Regression	3	3050	1016.67
Residual Error	26	2200	84.62
Total	29	5250	
Predictor	Coefficient	SE Coefficient	t
Constant	14.00	7.00	2.00
x_1	-1.00	0.70	-1.43
x_2	30.00	5.20	5.77
x_3	0.20	0.08	2.50

a. What are the estimated sales for the Bryne store, which has four competitors, a regional population of 0.4 (400,000), and an advertising expense of 30 ($30,000)?
b. Compute the R^2 value.
c. Compute the multiple standard error of estimate.
d. Conduct a global test of hypothesis to determine whether any of the regression coefficients are not equal to zero. Use the .05 level of significance.
e. Conduct tests of hypothesis to determine which of the independent variables have significant regression coefficients. Which variables would you consider eliminating? Use the .05 significance level.

18. Suppose that the sales manager of a large automotive parts distributor wants to estimate the total annual sales for each of the company's regions. Five factors appear to be related to regional sales: the number of retail outlets in the region, the number of automobiles in the region registered as of April 1, the total personal income recorded in the first quarter of the year, the average age of the automobiles (years), and the number of sales supervisors in the region. The data for each region were gathered for last year. For example, see the following table. In region 1 there were 1,739 retail outlets stocking the company's automotive parts, there were 9,270,000 registered automobiles in the region as of April 1, and so on. The region's sales for that year were $37,702,000.

Annual Sales ($ millions), y	Number of Retail Outlets, x_1	Number of Automobiles Registered (millions), x_2	Personal Income ($ billions), x_3	Average Age of Automobiles (years), x_4	Number of Supervisors, x_5
37.702	1,739	9.27	85.4	3.5	9.0
24.196	1,221	5.86	60.7	5.0	5.0
32.055	1,846	8.81	68.1	4.4	7.0
3.611	120	3.81	20.2	4.0	5.0
17.625	1,096	10.31	33.8	3.5	7.0
45.919	2,290	11.62	95.1	4.1	13.0
29.600	1,687	8.96	69.3	4.1	15.0
8.114	241	6.28	16.3	5.9	11.0
20.116	649	7.77	34.9	5.5	16.0
12.994	1,427	10.92	15.1	4.1	10.0

a. Consider the following correlation matrix. Which single variable has the strongest correlation with the dependent variable? The correlations between the independent variables outlets and income and between outlets and number of automobiles are fairly strong. Could this be a problem? What is this condition called?

	sales	outlets	cars	income	age
outlets	0.899				
automobiles	0.605	0.775			
income	0.964	0.825	0.409		
age	−0.323	−0.489	−0.447	−0.349	
bosses	0.286	0.183	0.395	0.155	0.291

b. The output for all five variables is shown below. What percent of the variation is explained by the regression equation?

```
The regression equation is
Sales = −19.7 − 0.00063 outlets + 1.74 autos + 0.410 income
        + 2.04 age − 0.034 bosses

           Predictor          Coef      SE Coef         T        P
           Constant        −19.672        5.422     −3.63    0.022
           outlets       −0.000629     0.002638     −0.24    0.823
           automobiles      1.7399       0.5530      3.15    0.035
           income          0.40994      0.04385      9.35    0.001
           age              2.0357       0.8779      2.32    0.081
           bosses          −0.0344       0.1880     −0.18    0.864

Analysis of Variance
           SOURCE            DF        SS        MS        F        P
           Regression         5   1593.81    318.76   140.36    0.000
           Residual Error     4      9.08      2.27
           Total              9   1602.89
```

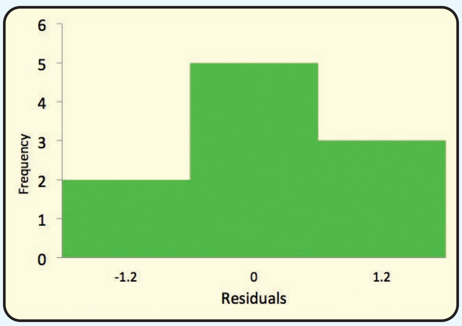
c. Conduct a global test of hypothesis to determine whether any of the regression coefficients are not zero. Use the .05 significance level.

d. Conduct a test of hypothesis on each of the independent variables. Would you consider eliminating “outlets” and “bosses”? Use the .05 significance level.

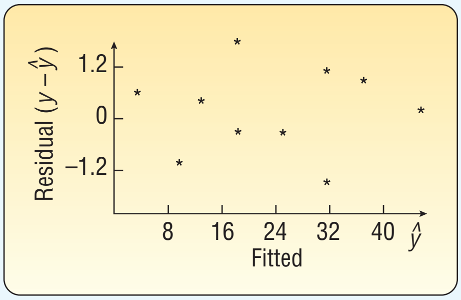
e. The regression has been rerun below with “outlets” and “bosses” eliminated. Compute the coefficient of determination. How much has R^2 changed from the previous analysis?

```
The regression equation is
Sales = −18.9 + 1.61 autos + 0.400 income + 1.96 age

           Predictor          Coef      SE Coef         T        P
           Constant        −18.924        3.636     −5.20    0.002
           automobiles      1.6129       0.1979      8.15    0.000
           income          0.40031      0.01569     25.52    0.000
           age              1.9637       0.5846      3.36    0.015


Analysis of Variance
           SOURCE            DF        SS        MS        F        P
           Regression         3   1593.66    531.22   345.25    0.000
           Residual Error     6      9.23      1.54
           Total              9   1602.89
```

f. Following is a histogram of the residuals. Does the normality assumption appear reasonable? Why?



g. Following is a plot of the fitted values of y (i.e., $\hat{y}$) and the residuals. What do you observe? Do you see any violations of the assumptions?



19. The administrator of a new paralegal program at Seagate Technical College wants to estimate the grade point average in the new program. He thought that high school GPA, the verbal score on the Scholastic Aptitude Test (SAT), and the mathematics score on the SAT would be good predictors of paralegal GPA. The data on nine students are:

Student	High School GPA	SAT Verbal	SAT Math	Paralegal GPA
1	3.25	480	410	3.21
2	1.80	290	270	1.68
3	2.89	420	410	3.58
4	3.81	500	600	3.92
5	3.13	500	490	3.00
6	2.81	430	460	2.82
7	2.20	320	490	1.65
8	2.14	530	480	2.30
9	2.63	469	440	2.33

a. Consider the following correlation matrix. Which variable has the strongest correlation with the dependent variable? Some of the correlations among the independent variables are strong. Does this appear to be a problem?

```
                 Paralegal   High School
                       GPA           GPA   SAT Verbal
High School GPA      0.911
SAT Verbal           0.616         0.609
SAT Math             0.487         0.636        0.599
```

b. Consider the following output. Compute the coefficient of multiple determination.

```
The regression equation is
Paralegal GPA = −0.411 + 1.20 HSGPA + 0.00163 SAT_Verbal − 0.00194
SAT_Math

Predictor          Coef        SE Coef         T          P
Constant        −0.4111         0.7823     −0.53      0.622
HSGPA            1.2014         0.2955      4.07      0.010
SAT_Verbal     0.001629       0.002147      0.76      0.482
SAT_Math      −0.001939       0.002074     −0.94      0.393

Analysis of Variance
SOURCE               DF        SS        MS         F         P
Regression            3    4.3595    1.4532     10.33     0.014
Residual Error        5    0.7036    0.1407
Total                 8    5.0631

SOURCE    DF    Seq SS
HSGPA      1    4.2061
SAT_Verbal 1    0.0303
SAT_Math   1     0.1231
```

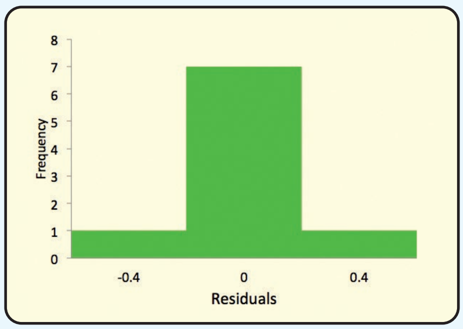
c. Conduct a global test of hypothesis from the preceding output. Does it appear that any of the regression coefficients are not equal to zero?

d. Conduct a test of hypothesis on each independent variable. Would you consider eliminating the variables "SAT_Verbal" and "SAT_Math"? Let $\alpha = .05$.

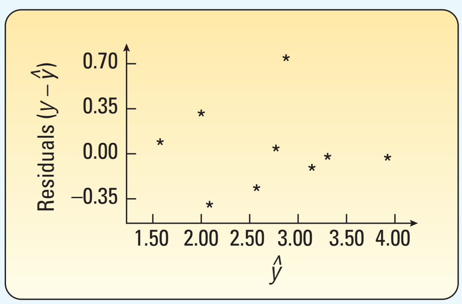
e. The analysis has been rerun without "SAT_Verbal" and "SAT_Math." See the following output. Compute the coefficient of determination. How much has R^2 changed from the previous analysis?

```
The regression equation is
Paralegal GPA = −0.454 + 1.16 HSGPA

Predictor        Coef     SE Coef        T        P
Constant      −0.4542      0.5542    −0.82    0.439
HSGPA          1.1589      0.1977     5.86    0.001

Analysis of Variance
SOURCE            DF        SS        MS        F        P
Regression         1    4.2061    4.2061    34.35    0.001
Residual Error     7    0.8570    0.1224
Total              8    5.0631
```

f. Following is a histogram of the residuals. Does the normality assumption for the residuals seem reasonable?



g. Following is a plot of the residuals and the $\hat{y}$ values. Do you see any violation of the assumptions?



20. **FILE** Mike Wilde is president of the teachers' union for Otsego School District. In preparing for upcoming negotiations, he is investigating the salary structure of classroom teachers in the district. He believes there are three factors that affect a teacher's salary: years of experience, a teaching effectiveness rating given by the principal, and whether the teacher has a master's degree. A random sample of 20 teachers resulted in the following data.

Salary ($ thousands), y	Years of Experience, x_1	Principal's Rating, x_2	Master's Degree,* x_3
31.1	8	35	0
33.6	5	43	0
29.3	2	51	1
⋮	⋮	⋮	⋮
30.7	4	62	0
32.8	2	80	1
42.8	8	72	0

*1 = yes, 0 = no.

a. Develop a correlation matrix. Which independent variable has the strongest correlation with the dependent variable? Does it appear there will be any problems with multicollinearity?
b. Determine the regression equation. What salary would you estimate for a teacher with 5 years' experience, a rating by the principal of 60, and no master's degree?
c. Conduct a global test of hypothesis to determine whether any of the regression coefficients differ from zero. Use the .05 significance level.
d. Conduct a test of hypothesis for the individual regression coefficients. Would you consider deleting any of the independent variables? Use the .05 significance level.
e. If your conclusion in part (d) was to delete one or more independent variables, run the analysis again without those variables.
f. Determine the residuals for the equation of part (e). Use a stem-and-leaf chart or a histogram to verify that the distribution of the residuals is approximately normal.
g. Plot the residuals computed in part (f) in a scatter diagram with the residuals on the Y-axis and the $\hat{y}$ values on the X-axis. Does the plot reveal any violations of the assumptions of regression?

21. **FILE** A video media consultant collected the following data on popular LED televisions sold through on-line retailers.

Manufacturer	Screen	Price	Manufacturer	Screen	Price
Sharp	46	736.50	Sharp	37	657.25
Samsung	52	1150.00	Sharp	32	426.75
Samsung	46	895.00	Sharp	52	1389.00
Sony	40	625.00	Samsung	40	874.75
Sharp	42	773.25	Sharp	32	517.50
Samsung	46	961.25	Samsung	52	1475.00
Samsung	40	686.00	Sony	40	954.25
Sharp	37	574.75	Sony	52	1551.50
Sharp	46	1000.00	Sony	46	1303.00
Sony	40	722.25	Sony	46	1430.50
Sony	52	1307.50	Sony	52	1717.00
Samsung	32	373.75			

a. Does there appear to be a linear relationship between the screen size and the price?
b. Which variable is the "dependent" variable?
c. Using statistical software, determine the regression equation. Interpret the value of the slope in the regression equation.
d. Include the manufacturer in a multiple linear regression analysis using a "dummy" variable. Does it appear that some manufacturers can command a premium price? Hint: You will need to use a set of indicator variables.
e. Test each of the individual coefficients to see if they are significant.
f. Make a plot of the residuals and comment on whether they appear to follow a normal distribution.
g. Plot the residuals versus the fitted values. Do they seem to have the same amount of variation?

22. **FILE** A regional planner is studying the demographics of nine counties in the eastern region of an Atlantic seaboard state. She has gathered the following data:

County	Median Income	Median Age	Coastal
A	$48,157	57.7	1
B	48,568	60.7	1
C	46,816	47.9	1
D	34,876	38.4	0
E	35,478	42.8	0
F	34,465	35.4	0
G	35,026	39.5	0
H	38,599	65.6	0
J	33,315	27.0	0

a. Is there a linear relationship between the median income and median age?
b. Which variable is the "dependent" variable?
c. Use statistical software to determine the regression equation. Interpret the value of the slope in a simple regression equation.
d. Include the aspect that the county is "coastal" or not in a multiple linear regression analysis using a "dummy" variable. Does it appear to be a significant influence on incomes?
e. Test each of the individual coefficients to see if they are significant.
f. Make a plot of the residuals and comment on whether they appear to follow a normal distribution.
g. Plot the residuals versus the fitted values. Do they seem to have the same amount of variation?

23. **FILE** Great Plains Distributors, Inc. sells roofing and siding products to home improvement retailers, such as Lowe's and Home Depot, and commercial contractors. The owner is interested in studying the effects of several variables on the sales volume of fiber-cement siding products.

The company has 26 marketing districts across the United States. In each district, it collected information on the following variables: sales volume (in thousands of dollars), advertising dollars (in thousands), number of active accounts, number of competing brands, and a rating of market potential.

Sales (000s)	Advertising Dollars (000s)	Number of Accounts	Number of Competitors	Market Potential
79.3	5.5	31	10	8
200.1	2.5	55	8	6
163.2	8.0	67	12	9
200.1	3.0	50	7	16
146.0	3.0	38	8	15
177.7	2.9	71	12	17
⋮	⋮	⋮	⋮	⋮
93.5	4.2	26	8	3
259.0	4.5	75	8	19
331.2	5.6	71	4	9

Conduct a multiple regression analysis to find the best predictors of sales.

a. Draw a scatter diagram comparing sales volume with each of the independent variables. Comment on the results.
b. Develop a correlation matrix. Do you see any problems? Does it appear there are any redundant independent variables?
c. Develop a regression equation. Conduct the global test. Can we conclude that some of the independent variables are useful in explaining the variation in the dependent variable?

d. Conduct a test of each of the independent variables. Are there any that should be dropped?
e. Refine the regression equation so the remaining variables are all significant.
f. Develop a histogram of the residuals and a normal probability plot. Are there any problems?
g. Determine the variance inflation factor for each of the independent variables. Are there any problems?

24. FILE A market researcher is studying on-line subscription services. She is particularly interested in what variables relate to the number of subscriptions for a particular on-line service. She is able to obtain the following sample information on 25 on-line subscription services. The following notation is used:

Sub = Number of subscriptions (in thousands)
Web page hits = Average monthly count (in thousands)
Adv = The advertising budget of the service (in $ hundreds)
Price = Average monthly subscription price ($)

Service	Sub	Web Page Hits	Adv	Price
1	37.95	588.9	13.2	35.1
2	37.66	585.3	13.2	34.7
3	37.55	566.3	19.8	34.8
⋮	⋮	⋮	⋮	⋮
23	38.83	629.6	22.0	35.3
24	38.33	680.0	24.2	34.7
25	40.24	651.2	33.0	35.8

a. Determine the regression equation.
b. Conduct a global test of hypothesis to determine whether any of the regression coefficients are not equal to zero.
c. Conduct a test for the individual coefficients. Would you consider deleting any coefficients?
d. Determine the residuals and plot them against the fitted values. Do you see any problems?
e. Develop a histogram of the residuals. Do you see any problems with the normality assumption?

25. FILE Fred G. Hire is the manager of human resources at Crescent Custom Steel Products. As part of his yearly report to the CEO, he is required to present an analysis of the salaried employees. For each of the 30 salaried employees, he records monthly salary; service at Crescent, in months; age; gender (1 = male, 0 = female); and whether the employee has a management or engineering position. Those employed in management are coded 0, and those in engineering are coded 1.

Sampled Employee	Monthly Salary	Length of Service	Age	Gender	Job
1	$1,769	93	42	1	0
2	1,740	104	33	1	0
3	1,941	104	42	1	1
⋮	⋮	⋮	⋮	⋮	⋮
28	1,791	131	56	0	1
29	2,001	95	30	1	1
30	1,874	98	47	1	0

a. Determine the regression equation, using salary as the dependent variable and the other four variables as independent variables.
b. What is the value of R^2? Comment on this value.
c. Conduct a global test of hypothesis to determine whether any of the independent variables are different from 0.

d. Conduct an individual test to determine whether any of the independent variables can be dropped.

e. Rerun the regression equation, using only the independent variables that are significant. How much more does a man earn per month than a woman? Does it make a difference whether the employee has a management or engineering position?

26. **FILE** Many regions in North and South Carolina and Georgia have experienced rapid population growth over the last 10 years. It is expected that the growth will continue over the next 10 years. This has motivated many of the large grocery store chains to build new stores in the region. The Kelley's Super Grocery Stores Inc. chain is no exception. The director of planning for Kelley's Super Grocery Stores wants to study adding more stores in this region. He believes there are two main factors that indicate the amount families spend on groceries. The first is their income and the other is the number of people in the family. The director gathered the following sample information.

Family	Food	Income	Size
1	$5.04	$73.98	4
2	4.08	54.90	2
3	5.76	94.14	4
⋮	⋮	⋮	⋮
23	4.56	38.16	3
24	5.40	43.74	7
25	4.80	48.42	5

Food and income are reported in thousands of dollars per year, and the variable size refers to the number of people in the household.

a. Develop a correlation matrix. Do you see any problems with multicollinearity?

b. Determine the regression equation. Discuss the regression equation. How much does an additional family member add to the amount spent on food?

c. What is the value of R^2? Can we conclude that this value is greater than 0?

d. Would you consider deleting either of the independent variables?

e. Plot the residuals in a histogram. Is there any problem with the normality assumption?

f. Plot the fitted values against the residuals. Does this plot indicate any problems with homoscedasticity?

27. **FILE** An investment advisor is studying the relationship between a common stock's price to earnings (P/E) ratio and factors that she thinks would influence it. She has the following data on the earnings per share (EPS) and the dividend percentage (Yield) for a sample of 20 stocks.

Stock	P/E	EPS	Yield
1	20.79	$2.46	1.42
2	3.03	2.69	4.05
3	44.46	−0.28	4.16
⋮	⋮	⋮	⋮
18	30.21	1.71	3.07
19	32.88	0.35	2.21
20	15.19	5.02	3.50

a. Develop a multiple linear regression with P/E as the dependent variable.

b. Are either of the two independent variables an effective predictor of P/E?

c. Interpret the regression coefficients.

d. Do any of these stocks look particularly undervalued?

e. Plot the residuals and check the normality assumption. Plot the fitted values against the residuals.

f. Does there appear to be any problems with homoscedasticity?

g. Develop a correlation matrix. Do any of the correlations indicate multicollinearity?

28. **FILE** The Conch Café, located in Gulf Shores, Alabama, features casual lunches with a great view of the Gulf of Mexico. To accommodate the increase in business during the summer vacation season, Fuzzy Conch, the owner, hires a large number of servers as seasonal help. When he interviews a prospective server, he would like to provide data on the amount a server can earn in tips. He believes that the amount of the bill and the number of diners are both related to the amount of the tip. He gathered the following sample information.

Customer	Amount of Tip	Amount of Bill	Number of Diners
1	$7.00	$48.97	5
2	4.50	28.23	4
3	1.00	10.65	1
⋮	⋮	⋮	⋮
28	2.50	26.25	2
29	9.25	56.81	5
30	8.25	50.65	5

a. Develop a multiple regression equation with the amount of tips as the dependent variable and the amount of the bill and the number of diners as independent variables. Write out the regression equation. How much does another diner add to the amount of the tips?

b. Conduct a global test of hypothesis to determine if at least one of the independent variables is significant. What is your conclusion?

c. Conduct an individual test on each of the variables. Should one or the other be deleted?

d. Use the equation developed in part (c) to determine the coefficient of determination. Interpret the value.

e. Plot the residuals. Is it reasonable to assume they follow the normal distribution?

f. Plot the residuals against the fitted values. Is it reasonable to conclude they are random?

29. **FILE** The president of Blitz Sales Enterprises sells kitchen products through cable television infomercials. He gathered data from the last 15 weeks of sales to determine the relationship between sales and the number of infomercials.

Infomercials	Sales ($000s)	Infomercials	Sales ($000s)
20	3.2	22	2.5
15	2.6	15	2.4
25	3.4	25	3.0
10	1.8	16	2.7
18	2.2	12	2.0
18	2.4	20	2.6
15	2.4	25	2.8
12	1.5		

a. Determine the regression equation. Are the sales predictable from the number of commercials?

b. Determine the residuals and plot a histogram. Does the normality assumption seem reasonable?

30. **FILE** The director of special events for Sun City believed that the amount of money spent on fireworks displays for the 4th of July was predictive of attendance at the Fall Festival held in October. She gathered the following data to test her suspicion.

4th of July ($000)	Fall Festival (000)	4th of July ($000)	Fall Festival (000)
10.6	8.8	9.0	9.5
8.5	6.4	10.0	9.8
12.5	10.8	7.5	6.6
9.0	10.2	10.0	10.1
5.5	6.0	6.0	6.1
12.0	11.1	12.0	11.3
8.0	7.5	10.5	8.8
7.5	8.4		

Determine the regression equation. Is the amount spent on fireworks related to attendance at the Fall Festival? Evaluate the regression assumptions by examining the residuals.

31. **FILE** You are a new hire at Laurel Woods Real Estate, which specializes in selling foreclosed homes via public auction. Your boss has asked you to use the following data (mortgage balance, monthly payments, payments made before default, and final auction price) on a random sample of recent sales in order to estimate what the actual auction price will be.

Loan	Monthly Payments	Payments Made	Auction Price
$ 85,600	$ 985.87	1	$16,900
115,300	902.56	33	75,800
103,100	736.28	6	43,900
⋮	⋮	⋮	⋮
119,400	1,021.23	58	69,000
90,600	836.46	3	35,600
104,500	1,056.37	22	63,000

a. Carry out a global test of hypothesis to verify if any of the regression coefficients are different from zero.
b. Do an individual test of the independent variables. Would you remove any of the variables?
c. If it seems one or more of the independent variables is not needed, remove it and work out the revised regression equation.

32. **FILE** Think about the figures from the previous exercise. Add a new variable that describes the potential interaction between the loan amount and the number of payments made. Then do a test of hypothesis to check if the interaction is significant.

DATA ANALYTICS

(The data for these exercises are available at the text website: **www.mhhe.com/Lind17e**.)

33. The North Valley Real Estate data reports information on homes on the market. Use the selling price of the home as the dependent variable and determine the regression equation using the size of the house, number of bedrooms, days on the market, and number of bathrooms as independent variables.
a. Develop a correlation matrix. Which independent variables have strong or weak correlations with the dependent variable? Do you see any problems with multicollinearity?
b. Use a statistical software package to determine the multiple regression equation. How did you select the variables to include in the equation? How did you use the information from the correlation analysis? Show that your regression equation shows a significant relationship. Write out the regression equation and interpret its practical application. Report and interpret the R-square.
c. Using your results from part (b), evaluate the addition of the variables: pool or garage. Report your results and conclusions.

d. Develop a histogram or a stem-and-leaf display of the residuals from the final regression equation developed in part (c). Is it reasonable to conclude that the normality assumption has been met?

e. Plot the residuals against the fitted values from the final regression equation developed in part (c). Plot the residuals on the vertical axis and the fitted values on the horizontal axis.

34. Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season. Let the number of games won be the dependent variable and the following variables be independent variables: team batting average, team Earned Run Average (ERA), number of home runs, and whether the team plays in the American or the National League.

a. Develop a correlation matrix. Which independent variables have strong or weak correlations with the dependent variable? Do you see any problems with multicollinearity? Are you surprised that the correlation coefficient for ERA is negative?

b. Use a statistical software package to determine the multiple regression equation. How did you select the variables to include in the equation? How did you use the information from the correlation analysis? Show that your regression equation shows a significant relationship. Write out the regression equation and interpret its practical application. Report and interpret the *R*-square. Is the number of wins affected by whether the team plays in the National or the American League?

c. Conduct a global test on the set of independent variables. Interpret.

d. Conduct a test of hypothesis on each of the independent variables. Would you consider deleting any of the variables? If so, which ones?

e. Develop a histogram or a stem-and-leaf display of the residuals from the final regression equation developed in part (f). Is it reasonable to conclude that the normality assumption has been met?

f. Plot the residuals against the fitted values from the final regression equation developed in part (f). Plot the residuals on the vertical axis and the fitted values on the horizontal axis.

35. Refer to the Lincolnville School District bus data. First, add a variable to change the type of engine (diesel or gasoline) to a qualitative variable. If the engine type is diesel, then set the qualitative variable to 0. If the engine type is gasoline, then set the qualitative variable to 1. Develop a regression equation using statistical software with maintenance cost as the dependent variable and age, odometer miles, miles since last maintenance, and engine type as the independent variables.

a. Develop a correlation matrix. Which independent variables have strong or weak correlations with the dependent variable? Do you see any problems with multicollinearity?

b. Use a statistical software package to determine the multiple regression equation. How did you select the variables to include in the equation? How did you use the information from the correlation analysis? Show that your regression equation shows a significant relationship. Write out the regression equation and interpret its practical application. Report and interpret the *R*-square.

c. Develop a histogram or a stem-and-leaf display of the residuals from the final regression equation developed in part (f). Is it reasonable to conclude that the normality assumption has been met?

d. Plot the residuals against the fitted values from the final regression equation developed in part (f) against the fitted values of *Y*. Plot the residuals on the vertical axis and the fitted values on the horizontal axis.

A REVIEW OF CHAPTERS 13–14

This section is a review of the major concepts and terms introduced in Chapters 13 and 14. Chapter 13 noted that the strength of the relationship between the independent variable and the dependent variable is measured by the *correlation coefficient.* The correlation coefficient is designated by the letter *r.* It can assume any value between −1.00 and +1.00 inclusive. Coefficients of −1.00 and +1.00 indicate a perfect relationship, and 0 indicates no relationship. A value near 0, such as −.14 or .14, indicates a weak relationship. A value near −1 or +1, such as −.90 or +.90, indicates a strong relationship. The *coefficient of determination,* also called R^2, measures the proportion of the total variation in the dependent variable explained by the independent variable. It can be computed as the square of the correlation coefficient.

Likewise, the strength of the relationship between several independent variables and a dependent variable is measured by the *coefficient of multiple determination,* R^2. It measures the proportion of the variation in y explained by two or more independent variables.

The linear relationship in the simple case involving one independent variable and one dependent variable is described by the equation $\hat{y} = a + bx$. For k independent variables, $x_1, x_2, \ldots x_k$, the same multiple regression equation is

$$\hat{y} = a + b_1x_1 + b_2x_2 + \ldots + b_kx_k$$

Solving for $b_1, b_2, \ldots, b_k$ would involve tedious calculations. Fortunately, this type of problem can be quickly solved using one of the many statistical software packages and spreadsheet packages. Various measures, such as the coefficient of determination, the multiple standard error of estimate, the results of the global test, and the test of the individual variables, are reported in the output of most statistical software programs.

PROBLEMS

1. The accounting department at Box and Go Apparel wishes to estimate the net profit for each of the chain's many stores on the basis of the number of employees in the store, overhead costs, average markup, and theft loss. The data from two stores are:

Store	Net Profit ($ thousands) $\hat{y}$	Number of Employees x_1	Overhead Cost ($ thousands) x_2	Average Markup (percent) x_3	Theft Loss ($ thousands) x_4
1	$846	143	$79	69%	$52
2	513	110	64	50	45

a. The dependent variable is __________.

b. The general equation for this problem is __________.

c. The multiple regression equation was computed to be $\hat{y} = 67 + 8x_1 - 10x_2 + 0.004x_3 - 3x_4$. What are the predicted sales for a store with 112 employees, an overhead cost of $65,000, a markup rate of 50%, and a loss from theft of $50,000?

d. Suppose R^2 was computed to be .86. Explain.

e. Suppose that the multiple standard error of estimate was 3 (in $ thousands). Explain what this means in this problem.

2. Quick-print firms in a large downtown business area spend most of their advertising dollars on displays on bus benches. A research project involves predicting monthly sales based on the annual amount spent on placing ads on bus benches. A sample of quick-print firms revealed these advertising expenses and sales:

Firm	Annual Bus Bench Advertising ($ thousands)	Monthly Sales ($ thousands)
A	2	10
B	4	40
C	5	30
D	7	50
E	3	20

a. Draw a scatter diagram.

b. Determine the correlation coefficient.

c. What is the coefficient of determination?

d. Compute the regression equation.

e. Estimate the monthly sales of a quick-print firm that spends $4,500 on bus bench advertisements.

f. Summarize your findings.

3. The following ANOVA output is given.

Analysis of Variance			
Source	DF	SS	MS
Regression	4	1050.8	262.70
Residual Error	20	83.8	4.19
Total	24	1134.6	
Predictor	Coefficient	SE Coefficient	t
Constant	70.06	2.13	32.89
x_1	0.42	0.17	2.47
x_2	0.27	0.21	1.29
x_3	0.75	0.30	2.50
x_4	0.42	0.07	6.00

a. Compute the coefficient of determination.
b. Compute the multiple standard error of estimate.
c. Conduct a test of hypothesis to determine whether any of the regression coefficients are different from zero.
d. Conduct a test of hypothesis on the individual regression coefficients. Can any of the variables be deleted?

CASES

A. The Century National Bank

Refer to the Century National Bank data. Using checking account balance as the dependent variable and using as independent variables the number of ATM transactions, the number of other services used, whether the individual has a debit card, and whether interest is paid on the particular account, write a report indicating which of the variables seem related to the account balance and how well they explain the variation in account balances. Should all of the independent variables proposed be used in the analysis or can some be dropped?

B. Terry and Associates: The Time to Deliver Medical Kits

Terry and Associates is a specialized medical testing center in Denver, Colorado. One of the firm's major sources of revenue is a kit used to test for elevated amounts of lead in the blood. Workers in auto body shops, those in the lawn care industry, and commercial house painters are exposed to large amounts of lead and thus must be randomly tested. It is expensive to conduct the test, so the kits are delivered on demand to a variety of locations throughout the Denver area.

Kathleen Terry, the owner, is concerned about setting appropriate costs for each delivery. To investigate, Ms. Terry gathered information on a random sample of 50 recent deliveries. Factors thought to be related to the cost of delivering a kit were:

Prep	The time in minutes between when the customized order is phoned into the company and when it is ready for delivery.
Delivery	The actual travel time in minutes from Terry's plant to the customer.
Mileage	The distance in miles from Terry's plant to the customer.

Sample Number	Cost	Prep	Delivery	Mileage	Sample Number	Cost	Prep	Delivery	Mileage
1	$32.60	10	51	20	6	$22.63	9	20	11
2	23.37	11	33	12	7	22.63	9	39	11
3	31.49	6	47	19	8	21.53	10	23	10
4	19.31	9	18	8	9	21.16	13	20	8
5	28.35	8	88	17	10	21.53	10	32	10

Sample Number	Cost	Prep	Delivery	Mileage	Sample Number	Cost	Prep	Delivery	Mileage
11	$28.17	5	35	16	31	$24.29	7	35	13
12	20.42	7	23	9	32	19.56	2	12	6
13	21.53	9	21	10	33	22.63	8	30	11
14	27.55	7	37	16	34	21.16	5	13	8
15	23.37	9	25	12	35	21.16	11	20	8
16	17.10	15	15	6	36	19.68	5	19	8
17	27.06	13	34	15	37	18.76	5	14	7
18	15.99	8	13	4	38	17.96	5	11	4
19	17.96	12	12	4	39	23.37	10	25	12
20	25.22	6	41	14	40	25.22	6	32	14
21	24.29	3	28	13	41	27.06	8	44	16
22	22.76	4	26	10	42	21.96	9	28	9
23	28.17	9	54	16	43	22.63	8	31	11
24	19.68	7	18	8	44	19.68	7	19	8
25	25.15	6	50	13	45	22.76	8	28	10
26	20.36	9	19	7	46	21.96	13	18	9
27	21.16	3	19	8	47	25.95	10	32	14
28	25.95	10	45	14	48	26.14	8	44	15
29	18.76	12	12	5	49	24.29	8	34	13
30	18.76	8	16	5	50	24.35	3	33	12

1. Develop a multiple linear regression equation that describes the relationship between the cost of delivery and the other variables. Do these three variables explain a reasonable amount of the variation in the dependent variable? Estimate the delivery cost for a kit that takes 10 minutes for preparation, takes 30 minutes to deliver, and must cover a distance of 14 miles.
2. Test to determine if one or more regression coefficient differs from zero. Also test to see whether any of the variables can be dropped from the analysis. If some of the variables can be dropped, rerun the regression equation until only significant variables are included.
3. Write a brief report interpreting the final regression equation.

PRACTICE TEST

Part 1—Objective

1. In a scatter diagram, the dependent variable is always scaled on which axis? 1. ________
2. What level of measurement is required to compute the correlation coefficient? 2. ________
3. If there is no correlation between two variables, what is the value of the correlation coefficient? 3. ________
4. Which of the following values indicates the strongest correlation between two variables? (.65, −.77, 0, −.12) 4. ________
5. Under what conditions will the coefficient of determination assume a value greater than 1? 5. ________

Given the following regression equation, $\hat{Y} = 7 - .5X$, and that the coefficient of determination is .81, answer questions 6, 7, and 8.

6. At what point does the regression equation cross the Y-axis? 6. ________
7. An increase of 1 unit in the independent variable will result in what amount of an increase or decrease in the dependent variable? 7. ________
8. What is the correlation coefficient? (Be careful of the sign.) 8. ________
9. If all the data points in a scatter diagram were on the regression line, what would be the value of the standard error of estimate? 9. ________
10. In a multiple regression equation, what is the maximum number of independent variables allowed? (2, 10, 30, unlimited) 10. ________

11. In multiple regression analysis, we assume what type of relationship between the dependent variable and the set of independent variables? (linear, multiple, curved, none of these) **11.** ________

12. The difference between Y and $\hat{Y}$ is called a ________. **12.** ________

13. For a dummy variable, such as gender, how many different values are possible? **13.** ________

14. What is the term given to a table that shows all possible correlation coefficients between the dependent variable and all the independent variables and among all the independent variables? **14.** ________

15. If there is a linear relationship between the dependent variable and the set of independent variables, a graph of the residuals will show what type of distribution? **15.** ________

Part 2—Problems

1. Given the following regression analysis output:

ANOVA Table

Source	SS	*df*	MS	*F*	*p*-value
Regression	129.7275	1	129.7275	14.50	.0007
Residual	250.4391	28	8.9443		
Total	380.1667	29			

Regression Output

Variables	Coefficients	Standard Error	*t* (*df* = 28)
Intercept	90.6190	1.5322	59.141
Slope	−0.9401	0.2468	−3.808

a. What is the sample size?
b. Write out the regression equation. Interpret the slope and intercept values.
c. If the value of the independent variable is 10, what is the value of the dependent variable?
d. Calculate the coefficient of determination. Interpret this value.
e. Calculate the correlation coefficient. Conduct a test of hypothesis to determine if there is a significant negative association between the variables.

2. Given the following regression analysis output.

ANOVA Table

Source	SS	*df*	MS	*F*	*p*-value
Regression	227.0928	4	56.7732	9.27	0.000
Residual	153.0739	25	6.1230		
Total	380.1667	29			

Regression Output

Variables	Coefficients	Standard Error	*t* (*df* = 25)	*p*-value
Intercept	68.3366	8.9752	7.614	0.000
x_1	0.8595	0.3087	2.784	0.010
x_2	−0.3380	0.8381	−0.403	0.690
x_3	−0.8179	0.2749	−2.975	0.006
x_4	−0.5824	0.2541	−2.292	0.030

a. What is the sample size?
b. How many independent variables are in the study?
c. Determine the coefficient of determination.
d. Conduct a global test of hypothesis. Can you conclude at least one of the independent variables does not equal zero? Use the .01 significance level.
e. Conduct an individual test of hypothesis on each of the independent variables. Would you consider dropping any of the independent variables? If so, which variable or variables would you drop? Use the .01 significance level.